SEMIPARAMETRIC DENSITY RATIO MODELS: ASYMPTOTIC THEORY, COMPUTATION, AND APPLICATIONS

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CONTENTS

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ABSTRACT. We present a systematic and rigorous framework for empirical likelihood-based inference in semiparametric density ratio models (DRMs). When the density ratio function is suitably specified, the DRM reduces to a class of multinomial regression models with commonly used link functions within the generalized linear model (GLM) framework, particularly under stratified sampling designs such as case-control studies. This equivalence is especially relevant for balanced case-control settings in clinical research and epidemiology, where it enables practitioners to leverage standard GLM software for semiparametric estimation.

We develop three complementary computational algorithms—nested optimization, joint Newton–Raphson, and an EM-type fixed-point iteration—each equipped with proven convergence guarantees. For the exponential tilt model, we derive closed-form expressions that eliminate inner root-finding and significantly reduce computational complexity. We further extend the framework to accommodate multiple populations, establishing existence, uniqueness, and asymptotic normality of the maximum empirical likelihood estimators.

Our asymptotic theory encompasses weak convergence of the estimated cumulative distribution functions to tight Gaussian processes, semiparametric efficiency bounds, and Wilks-type theorems for likelihood ratio testing. We also discuss extensions of the DRM framework to contemporary machine learning problems, including rare event learning, anomaly detection, and novelty detection. These results provide a solid theoretical foundation for both future algorithmic developments in machine learning and practical applications in data science.

1. INTRODUCTION

The density ratio model (DRM) has emerged as a powerful and flexible semiparametric framework for comparing two or more populations. Its conceptual origins can be traced to early work on logistic discrimination by Anderson [1972, 1979] and to the theory of biased sampling developed by Vardi [1982, 1985]. However, the model was formally introduced and explicitly named the *density ratio model* by Qin [1998], whose seminal *Biometrika* paper established the fundamental connection between case-control studies and semiparametric two-sample density ratio models. In this framework, the log-density ratio between two populations is assumed to take a known parametric form, while the baseline density remains completely unspecified. This formulation was subsequently extended and popularized through the work of Qin and Zhang [1997], who integrated the model with empirical likelihood methods, and by Fokianos and Troendle [2007], who applied it to inference for relative treatment effects.

Density ratio models provide a compelling compromise between the flexibility of fully nonparametric methods and the efficiency of parametric inference. By specifying only the ratio of density functions while leaving the baseline distribution unspecified, the DRM captures distributional differences through a finite-dimensional parameter without imposing restrictive assumptions on the underlying data-generating process. This class of models has attracted substantial interest in biostatistics, epidemiology, econometrics, and, more recently, machine learning, owing to its ability to pool information across samples and yield estimators with improved asymptotic efficiency.

The seminal work of Qin [1998] introduced the empirical likelihood approach to the two-sample DRM with an exponential tilt weight function of the form $w(x) = \exp(\alpha + \boldsymbol{\beta}^\top \mathbf{r}(x))$, motivated by the binary logistic regression model under stratified sampling (also called case-control sampling in clinical settings). This exponential tilt model has been widely applied in case-control studies Prentice and Pyke [1979], receiver operating characteristic (ROC) curve analysis Qin and Zhang [2003], Wan and Zhang [2017], quantile estimation Chen and Liu [2013], and missing data problems Qin et al. [2002]. The empirical likelihood methodology, originally developed by Owen [1988, 1990, 2001], offers a powerful nonparametric likelihood framework that automatically incorporates parameter constraints and yields Wilks-type theorems without requiring explicit specification of the underlying distribution.

Despite the considerable progress in DRM methodology, several important gaps remain in the literature. First, most existing treatments focus on the exponential tilt specification, leaving a general theory for arbitrary density ratio functions underdeveloped. Second, while computational algorithms have been proposed, rigorous convergence guarantees and comparative analyses remain scarce. Third, the extension to multiple populations—essential for multi-class classification and multi-arm clinical trials—lacks a unified theoretical treatment. Fourth, although the connections between DRMs and generalized linear models have been recognized in special cases, they have not been fully exploited for computational and practical advantage.

This paper addresses these gaps by developing a comprehensive theoretical and methodological framework for *generalized* density ratio models. Our contributions are organized as follows:

(1) We formulate a general DRM that accommodates arbitrary positive weight functions $w(x;\boldsymbol{\gamma})$ subject only to normalization constraints, and derive the maximum empirical likelihood estimators (MELEs) for both the parameter vector $\boldsymbol{\gamma}$ and the baseline distribution G_0.
(2) We develop three complementary computational strategies—nested optimization, joint Newton–Raphson, and an EM-type fixed-point iteration—each tailored to different problem structures and initial value regimes. We prove convergence guarantees for each algorithm under standard regularity conditions.
(3) For the exponential tilt model, we derive closed-form expressions for the Lagrange multipliers, thereby eliminating inner root-finding, reducing computational complexity by an order of magnitude, and establishing the precise equivalence to multinomial logistic regression within the GLM framework.
(4) We extend the theory to $K \geq 2$ populations, proving existence and uniqueness of the MELEs, deriving the profile score and information matrix, and establishing asymptotic normality under multi-sample sampling designs.
(5) We develop the asymptotic theory for the two-sample case, including weak convergence of the estimated cumulative distribution functions to tight Gaussian processes, characterization of the semiparametric efficiency bound, and Wilks-type theorems for likelihood ratio, score, and Wald tests.
(6) We discuss extensions of the DRM framework to contemporary machine learning problems, including rare event learning, anomaly detection, and novelty detection, highlighting how the DRM provides a principled statistical foundation for these applications.

The remainder of the paper is organized as follows. Section 2 introduces the general DRM and derives the maximum empirical likelihood estimation framework. Section 3 presents the three computational algorithms and establishes their convergence properties. Section 4 specializes the theory to the exponential tilt model and establishes the GLM equivalence. Section 5 extends the framework to multiple populations. Section 6 develops the asymptotic theory for the two-sample case, including consistency, normality, and weak convergence. Section 7 characterizes the semiparametric efficiency bound. Section 8 applies these results to hypothesis testing. Sections 9 and 10 discuss extensions to rare-event detection and survey broader application domains. Section 11 concludes with a discussion of future directions.

2. General Density Ratio Model

Let $G_0(x)$ and $G_1(x)$ denote the cumulative distribution functions (CDFs) of two independent populations with corresponding density functions $g_0(x)$ and $g_1(x)$, respectively. We consider the general density ratio model

$$\frac{g_1(x)}{g_0(x)} = w(x;\boldsymbol{\gamma}), \tag{2.1}$$

where $w(x;\boldsymbol{\gamma})$ is a specified density ratio function with parameter vector $\boldsymbol{\gamma} = (\gamma_1, \gamma_2, \ldots, \gamma_p)^\top \in \Theta \subset \mathbb{R}^p$. The function $w(x;\boldsymbol{\gamma})$ satisfies the following conditions:

- (C1) **Positivity:** $w(x;\boldsymbol{\gamma}) > 0$ for all $x \in \mathscr{D}$ and all $\boldsymbol{\gamma} \in \Theta$;
- (C2) **Normalization:** $\int_{\mathscr{D}} w(x;\boldsymbol{\gamma})\, dG_0(x) = 1$, where $\mathscr{D}$ is the common support of $g_0(x)$ and $g_1(x)$.

Condition (C2) ensures that $g_1(x) = w(x;\boldsymbol{\gamma})g_0(x)$ is a proper probability density function. We observe two independent random samples

$$\begin{aligned} x_1, x_2, \ldots, x_{n_0} &\overset{\text{i.i.d.}}{\sim} g_0(x), \\ z_1, z_2, \ldots, z_{n_1} &\overset{\text{i.i.d.}}{\sim} g_1(x) = w(x;\boldsymbol{\gamma})g_0(x). \end{aligned} \tag{2.2}$$

For notational convenience, we write $w(x) \equiv w(x;\boldsymbol{\gamma})$ when the parameter dependence is clear from context.

Remark 1. The density ratio constraint (2.1) does not imply that the two populations are correlated; rather, it specifies that their distributional difference is captured entirely by the parametric weight function $w(x;\boldsymbol{\gamma})$. The baseline density $g_0(x)$ remains completely unspecified, rendering the model semiparametric.

2.1. **Maximum Empirical Likelihood Estimation.** To estimate the unknown parameter $\boldsymbol{\gamma}$ and the baseline distribution G_0, we combine the two samples into a pooled dataset

$$\{t_1, t_2, \ldots, t_n\} = \{x_1, x_2, \ldots, x_{n_0}, z_1, z_2, \ldots, z_{n_1}\}, \tag{2.3}$$

where $n = n_0 + n_1$. Let $p_i = dG_0(t_i)$ denote the probability mass at t_i under the baseline distribution, so that $q_i = dG_1(t_i) = p_i w(t_i;\boldsymbol{\gamma})$ for $i = 1, 2, \ldots, n$. The semiparametric empirical likelihood function for model (2.1) is

$$\mathscr{L}(\boldsymbol{\gamma}, \mathbf{p}) = \prod_{i=1}^{n_0} dG_0(x_i) \prod_{j=1}^{n_1} dG_1(z_j) = \prod_{i=1}^{n} p_i \prod_{j=1}^{n_1} w(z_j;\boldsymbol{\gamma}), \tag{2.4}$$

subject to the constraints

$$\sum_{i=1}^{n} p_i = 1, \qquad \sum_{i=1}^{n} p_i\big[w(t_i;\boldsymbol{\gamma}) - 1\big] = 0, \qquad p_i \geq 0. \tag{2.5}$$

The first constraint ensures that $\{p_i\}$ forms a probability distribution; the second enforces the DRM normalization condition (C2); the nonnegativity constraint is implicit in the likelihood formulation.

The corresponding log-likelihood function is

$$\ell(\boldsymbol{\gamma}, \mathbf{p}) = \sum_{i=1}^{n} \log p_i + \sum_{j=1}^{n_1} \log w(z_j;\boldsymbol{\gamma}), \tag{2.6}$$

subject to the same constraints (2.5).

To maximize (2.6) subject to (2.5), we form the Lagrangian

$$Q(\boldsymbol{\gamma}, \mathbf{p}, \lambda_0, \lambda_1) = \sum_{i=1}^{n} \log p_i + \sum_{j=1}^{n_1} \log w(z_j; \boldsymbol{\gamma}) - \lambda_0 \left(\sum_{i=1}^{n} p_i - 1 \right) - \lambda_1 \sum_{i=1}^{n} p_i \big[w(t_i; \boldsymbol{\gamma}) - 1 \big]. \tag{2.7}$$

Differentiating with respect to p_i and setting the result to zero yields

$$\frac{\partial Q}{\partial p_i} = \frac{1}{p_i} - \lambda_0 - \lambda_1 \big[w(t_i; \boldsymbol{\gamma}) - 1 \big] = 0. \tag{2.8}$$

Solving for p_i, we obtain

$$p_i = \frac{1}{\lambda_0 + \lambda_1 \big[w(t_i; \boldsymbol{\gamma}) - 1 \big]}. \tag{2.9}$$

Summing (2.9) over $i = 1, \ldots, n$ and applying the normalization constraint yields the analytical solution $\lambda_0 = n$. Substituting back, the baseline probabilities become

$$p_i = \frac{1}{n + \lambda_1 \big[w(t_i; \boldsymbol{\gamma}) - 1 \big]}, \qquad i = 1, \ldots, n. \tag{2.10}$$

Substituting (2.10) into the second constraint of (2.5) yields the implicit equation

$$\Phi(\lambda_1; \boldsymbol{\gamma}) \equiv \sum_{i=1}^{n} \frac{w(t_i; \boldsymbol{\gamma}) - 1}{n + \lambda_1 \big[w(t_i; \boldsymbol{\gamma}) - 1 \big]} = 0. \tag{2.11}$$

Proposition 1 (Implicit Function for λ_1)**.** *For fixed $\boldsymbol{\gamma}$, equation (2.11) defines λ_1 as an implicit function $\lambda_1(\boldsymbol{\gamma})$, provided $w(t_i; \boldsymbol{\gamma}) \neq 1$ for at least one i and the denominator does not vanish for any i.*

Proof. For each i, the term $\big[w(t_i; \boldsymbol{\gamma}) - 1 \big] / \big\{ n + \lambda_1 \big[w(t_i; \boldsymbol{\gamma}) - 1 \big] \big\}$ is strictly monotone in λ_1 (decreasing if $w > 1$, increasing if $w < 1$). Hence $\Phi(\cdot; \boldsymbol{\gamma})$ is strictly monotone on any interval avoiding poles $\lambda_1 = -n / \big[w(t_i; \boldsymbol{\gamma}) - 1 \big]$. By the intermediate value theorem, a unique root exists in the appropriate interval. $\square$

2.2. Profile Log-Likelihood.

Substituting (2.10) into (2.6) yields the *profile log-likelihood*

$$\ell_{\text{prof}}(\boldsymbol{\gamma}) = \sum_{j=1}^{n_1} \log w(z_j; \boldsymbol{\gamma}) - \sum_{i=1}^{n} \log \Big\{ n + \lambda_1(\boldsymbol{\gamma}) \big[w(t_i; \boldsymbol{\gamma}) - 1 \big] \Big\} - n \log n, \tag{2.12}$$

where $\lambda_1(\boldsymbol{\gamma})$ is the unique solution to (2.11).

Theorem 2 (Well-Posedness of the MELE)**.** *Under regularity conditions on $w(x; \boldsymbol{\gamma})$—positivity, twice continuous differentiability, and finite moments—the maximum empirical likelihood estimator $\hat{\boldsymbol{\gamma}} = \arg\max_{\boldsymbol{\gamma} \in \Theta} \ell_{\text{prof}}(\boldsymbol{\gamma})$ exists and is consistent for the true parameter $\boldsymbol{\gamma}_0$.*

2.3. **Gradient of the Profile Log-Likelihood.** To apply gradient-based optimization, we require $\nabla_{\boldsymbol{\gamma}}\ell_{\text{prof}}(\boldsymbol{\gamma})$. By the chain rule,

$$\nabla_{\boldsymbol{\gamma}}\ell_{\text{prof}}(\boldsymbol{\gamma}) = \underbrace{\sum_{j=1}^{n_1}\frac{\nabla_{\boldsymbol{\gamma}}w(z_j;\boldsymbol{\gamma})}{w(z_j;\boldsymbol{\gamma})}}_{\text{case-sample contribution}} - \underbrace{\sum_{i=1}^{n}\frac{\lambda_1(\boldsymbol{\gamma})\nabla_{\boldsymbol{\gamma}}w(t_i;\boldsymbol{\gamma}) + \big[w(t_i;\boldsymbol{\gamma})-1\big]\nabla_{\boldsymbol{\gamma}}\lambda_1(\boldsymbol{\gamma})}{n+\lambda_1(\boldsymbol{\gamma})\big[w(t_i;\boldsymbol{\gamma})-1\big]}}_{\text{baseline contribution}}. \tag{2.13}$$

The term $\nabla_{\boldsymbol{\gamma}}\lambda_1(\boldsymbol{\gamma})$ is obtained by implicit differentiation of (2.11). Differentiating $\Phi(\lambda_1(\boldsymbol{\gamma});\boldsymbol{\gamma}) = 0$ gives

$$\frac{\partial\Phi}{\partial\lambda_1}\nabla_{\boldsymbol{\gamma}}\lambda_1 + \nabla_{\boldsymbol{\gamma}}\Phi = \mathbf{0}, \tag{2.14}$$

where

$$\frac{\partial\Phi}{\partial\lambda_1} = -\sum_{i=1}^{n}\frac{\big[w(t_i;\boldsymbol{\gamma})-1\big]^2}{\big\{n+\lambda_1(\boldsymbol{\gamma})\big[w(t_i;\boldsymbol{\gamma})-1\big]\big\}^2}, \tag{2.15}$$

$$\nabla_{\boldsymbol{\gamma}}\Phi = \sum_{i=1}^{n}\frac{n\,\nabla_{\boldsymbol{\gamma}}w(t_i;\boldsymbol{\gamma})}{\big\{n+\lambda_1(\boldsymbol{\gamma})\big[w(t_i;\boldsymbol{\gamma})-1\big]\big\}^2}. \tag{2.16}$$

Therefore,

$$\nabla_{\boldsymbol{\gamma}}\lambda_1(\boldsymbol{\gamma}) = -\frac{\nabla_{\boldsymbol{\gamma}}\Phi}{\partial\Phi/\partial\lambda_1} = \frac{\displaystyle\sum_{i=1}^{n}\frac{n\,\nabla_{\boldsymbol{\gamma}}w(t_i;\boldsymbol{\gamma})}{\big\{n+\lambda_1\big[w(t_i;\boldsymbol{\gamma})-1\big]\big\}^2}}{\displaystyle\sum_{i=1}^{n}\frac{\big[w(t_i;\boldsymbol{\gamma})-1\big]^2}{\big\{n+\lambda_1\big[w(t_i;\boldsymbol{\gamma})-1\big]\big\}^2}}. \tag{2.17}$$

In practice, when analytical derivatives of w are unavailable or cumbersome, numerical differentiation via finite differences provides a robust alternative.

3. Computational Algorithms

We now develop three complementary algorithms for computing the maximum empirical likelihood estimators. Each algorithm is suited to different problem characteristics: nested optimization for general weight functions, joint Newton–Raphson for problems where analytical or numerical Hessians are available, and an EM-type iteration for settings with poor initial values or irregular likelihood surfaces.

3.1. **Algorithm 1: Nested Optimization.** The nested optimization approach separates the problem into an outer loop over $\boldsymbol{\gamma}$ and an inner root-finding for $\lambda_1(\boldsymbol{\gamma})$. This is the most robust strategy when analytical derivatives are unavailable or the weight function is complex.
Computational Complexity. Each profile evaluation requires one root-finding call costing $O(n)$ operations. The BFGS optimizer typically converges in $O(p^2)$ iterations, yielding overall complexity $O(p^2 \cdot n \cdot N_{\text{root}})$, where N_{root} denotes the average number of bisection steps.

Algorithm 1 Nested Optimization for General DRM

Require: Combined sample $\mathbf{t} = (t_1, \ldots, t_n)$; case sample $\mathbf{z} = (z_1, \ldots, z_{n_1})$; weight function $w(x; \boldsymbol{\gamma})$; gradient $\nabla_{\boldsymbol{\gamma}} w(x; \boldsymbol{\gamma})$; initial value $\boldsymbol{\gamma}^{(0)}$; tolerance ε.

Ensure: MELE $\hat{\boldsymbol{\gamma}}$; Lagrange multiplier $\hat{\lambda}_1$; baseline masses $\hat{p}_i$.

Inner Subroutine: SOLVELAMBDA1$(\boldsymbol{\gamma}, \mathbf{t}, n)$

Define $\Phi(\lambda_1) \leftarrow \sum_{i=1}^{n} \frac{w(t_i;\boldsymbol{\gamma})-1}{n+\lambda_1\left[w(t_i;\boldsymbol{\gamma})-1\right]}$
$\lambda_1^{\text{low}} \leftarrow -n \big/ \big(\max_i w(t_i; \boldsymbol{\gamma}) - 1 + \delta\big)$ ▷ $\delta > 0$ small
$\lambda_1^{\text{high}} \leftarrow -n \big/ \big(\min_i w(t_i; \boldsymbol{\gamma}) - 1 - \delta\big)$
while $\Phi(\lambda_1^{\text{low}}) \cdot \Phi(\lambda_1^{\text{high}}) > 0$ **do**
    $\lambda_1^{\text{low}} \leftarrow 2\lambda_1^{\text{low}}; \quad \lambda_1^{\text{high}} \leftarrow 2\lambda_1^{\text{high}}$
end while
$\lambda_1 \leftarrow$ UNIROOT$\big(\Phi, [\lambda_1^{\text{low}}, \lambda_1^{\text{high}}]\big)$
return λ_1

Outer Optimization:

function PROFILELOGLIKELIHOOD$(\boldsymbol{\gamma})$
    $\mathbf{w}_t \leftarrow \big(w(t_1; \boldsymbol{\gamma}), \ldots, w(t_n; \boldsymbol{\gamma})\big)$
    $\mathbf{w}_z \leftarrow \big(w(z_1; \boldsymbol{\gamma}), \ldots, w(z_{n_1}; \boldsymbol{\gamma})\big)$
    if $\min(\mathbf{w}_t) \leq 0$ **or** $\min(\mathbf{w}_z) \leq 0$ **then**
        return $-\infty$
    end if
    $\lambda_1 \leftarrow$ SOLVELAMBDA1$(\boldsymbol{\gamma}, \mathbf{t}, n)$
    $\ell \leftarrow \sum_{j=1}^{n_1} \log w(z_j; \boldsymbol{\gamma}) - \sum_{i=1}^{n} \log\big(n + \lambda_1\left[w(t_i; \boldsymbol{\gamma}) - 1\right]\big)$
    return ℓ
end function
$\hat{\boldsymbol{\gamma}} \leftarrow \arg\max_{\boldsymbol{\gamma}}$ PROFILELOGLIKELIHOOD$(\boldsymbol{\gamma})$ ▷ BFGS or L-BFGS-B
$\hat{\lambda}_1 \leftarrow$ SOLVELAMBDA1$(\hat{\boldsymbol{\gamma}}, \mathbf{t}, n)$
$\hat{p}_i \leftarrow \left[n + \hat{\lambda}_1\big(w(t_i; \hat{\boldsymbol{\gamma}}) - 1\big)\right]^{-1}, \quad i = 1, \ldots, n$
return $(\hat{\boldsymbol{\gamma}}, \hat{\lambda}_1, \hat{p}_1, \ldots, \hat{p}_n)$

Remarks. Algorithm 1 is the most robust approach for general weight functions. The inner root-finding is guaranteed to converge by Proposition 1. Numerical gradients (finite differences) may be used if analytical derivatives of w are unavailable.

3.2. **Algorithm 2: Joint Newton–Raphson.**

The joint Newton–Raphson approach treats $(\boldsymbol{\gamma}, \lambda_1)$ as a $(p+1)$-dimensional parameter and solves the augmented score system simultaneously. This achieves quadratic convergence near the optimum when analytical or numerical Hessians are available.

Computational Complexity. Each iteration costs $O((p+1)^2 \cdot n)$ for Hessian construction plus $O((p+1)^3)$ for the linear solve. Near the optimum, convergence is quadratic.

Algorithm 2 Joint Newton–Raphson for Augmented System

Require: Combined sample $\mathbf{t}$; case sample $\mathbf{z}$; weight function $w(x;\boldsymbol{\gamma})$; gradient $\nabla_{\boldsymbol{\gamma}} w(x;\boldsymbol{\gamma})$; initial $\boldsymbol{\gamma}^{(0)}$; tolerance ε; max iterations M.
Ensure: MELE $\hat{\boldsymbol{\gamma}}$; $\hat{\lambda}_1$; baseline masses $\hat{p}_i$.

Initialization:
1: $\boldsymbol{\gamma} \leftarrow \boldsymbol{\gamma}^{(0)}$
2: $\mathbf{w}_t \leftarrow \big(w(t_1;\boldsymbol{\gamma}),\dots,w(t_n;\boldsymbol{\gamma})\big)$
3: $\lambda_1 \leftarrow \text{SOLVELAMBDA1}(\boldsymbol{\gamma},\mathbf{t},n)$ ▷ Warm start via Algorithm 1

Main Iteration:
4: **for** $m = 1,2,\dots,M$ **do**
5: $\quad\mathbf{w}_t \leftarrow \big(w(t_1;\boldsymbol{\gamma}),\dots,w(t_n;\boldsymbol{\gamma})\big);\quad \mathbf{w}_z \leftarrow \big(w(z_1;\boldsymbol{\gamma}),\dots,w(z_{n_1};\boldsymbol{\gamma})\big)$
6: $\quad\nabla\mathbf{w}_t \leftarrow [\nabla_{\boldsymbol{\gamma}} w(t_i;\boldsymbol{\gamma})]_{i=1}^{n}$ ▷ $n \times p$ matrix
7: $\quad\nabla\mathbf{w}_z \leftarrow [\nabla_{\boldsymbol{\gamma}} w(z_j;\boldsymbol{\gamma})]_{j=1}^{n_1}$ ▷ $n_1 \times p$ matrix
8: $\quad d_i \leftarrow n + \lambda_1\big[w(t_i;\boldsymbol{\gamma}) - 1\big],\quad i = 1,\dots,n$

Compute Augmented Score:
9: $\quad\mathbf{S}_{\boldsymbol{\gamma}} \leftarrow \sum_{j=1}^{n_1} \frac{\nabla_{\boldsymbol{\gamma}} w(z_j;\boldsymbol{\gamma})}{w(z_j;\boldsymbol{\gamma})} - \sum_{i=1}^{n} \frac{\lambda_1 \nabla_{\boldsymbol{\gamma}} w(t_i;\boldsymbol{\gamma})}{d_i}$ ▷ $p \times 1$
10: $\quad S_{\lambda_1} \leftarrow \sum_{i=1}^{n} \frac{w(t_i;\boldsymbol{\gamma}) - 1}{d_i}$ ▷ scalar
11: $\quad\mathbf{S} \leftarrow (\mathbf{S}_{\boldsymbol{\gamma}}^\top, S_{\lambda_1})^\top$ ▷ $(p+1)\times 1$
12: $\quad$**if** $\|\mathbf{S}\|_2 < \varepsilon$ **then break** ▷ Convergence achieved
13: $\quad$**end if**

Compute Numerical Hessian $\mathbf{H} \in \mathbb{R}^{(p+1)\times(p+1)}$:
14: $\quad$**for** $k = 1,\dots,p+1$ **do**
15: $\qquad\boldsymbol{\theta} \leftarrow (\boldsymbol{\gamma}^\top, \lambda_1)^\top;\quad \boldsymbol{\theta}^{(+)} \leftarrow \boldsymbol{\theta} + \varepsilon_{\text{hess}} \cdot \mathbf{e}_k$
16: $\qquad$Compute $\mathbf{S}^{(+)}$ at $\boldsymbol{\theta}^{(+)}$ using steps 8–11
17: $\qquad\mathbf{H}_{\cdot k} \leftarrow (\mathbf{S}^{(+)} - \mathbf{S})/\varepsilon_{\text{hess}}$
18: $\quad$**end for**
19: $\quad\mathbf{H} \leftarrow (\mathbf{H} + \mathbf{H}^\top)/2$ ▷ Symmetrize

Newton Step with Line Search:
20: $\quad\boldsymbol{\Delta} \leftarrow -\mathbf{H}^{-1}\mathbf{S}$ ▷ Solve $\mathbf{H}\boldsymbol{\Delta} = -\mathbf{S}$
21: $\quad\alpha \leftarrow 1$
22: $\quad$**for** $\ell = 1,\dots,20$ **do**
23: $\qquad\boldsymbol{\theta}^{\text{new}} \leftarrow \boldsymbol{\theta} + \alpha\boldsymbol{\Delta}$
24: $\qquad\boldsymbol{\gamma}^{\text{new}} \leftarrow \boldsymbol{\theta}^{\text{new}}_{1:p};\quad \lambda_1^{\text{new}} \leftarrow \boldsymbol{\theta}^{\text{new}}_{p+1}$
25: $\qquad$**if** $\min_i w(t_i;\boldsymbol{\gamma}^{\text{new}}) > 0$ **and** $\min_i d_i^{\text{new}} > 0$ **then break**
26: $\qquad$**end if**
27: $\qquad\alpha \leftarrow \alpha/2$
28: $\quad$**end for**
29: $\quad\boldsymbol{\gamma} \leftarrow \boldsymbol{\gamma}^{\text{new}};\quad \lambda_1 \leftarrow \lambda_1^{\text{new}}$
30: **end for**

Finalization:
31: $\hat{\boldsymbol{\gamma}} \leftarrow \boldsymbol{\gamma};\quad \hat{\lambda}_1 \leftarrow \lambda_1$
32: $\hat{p}_i \leftarrow \big[n + \hat{\lambda}_1\big(w(t_i;\hat{\boldsymbol{\gamma}}) - 1\big)\big]^{-1},\quad i = 1,\dots,n$
33: **return** $(\hat{\boldsymbol{\gamma}}, \hat{\lambda}_1, \hat{p}_1,\dots,\hat{p}_n)$

3.3. **Algorithm 3: EM-Type Fixed-Point Iteration.** The EM-type algorithm alternates between solving λ_1 given $\boldsymbol{\gamma}$ (E-step) and optimizing $\boldsymbol{\gamma}$ given λ_1 (M-step). This is the most stable approach when initial values are poor or the likelihood surface is irregular.

Algorithm 3 EM-Type Algorithm for General DRM

Require: Combined sample $\mathbf{t}$; case sample $\mathbf{z}$; weight function $w(x;\boldsymbol{\gamma})$; gradient $\nabla_{\boldsymbol{\gamma}} w(x;\boldsymbol{\gamma})$; initial $\boldsymbol{\gamma}^{(0)}$; tolerance ε; max iterations M.
Ensure: MELE $\hat{\boldsymbol{\gamma}}$; $\hat{\lambda}_1$; baseline masses $\hat{p}_i$.

Initialization:
$\boldsymbol{\gamma} \leftarrow \boldsymbol{\gamma}^{(0)}; \quad \ell_{\text{old}} \leftarrow -\infty$

Main Iteration:
for $m = 1, 2, \ldots, M$ **do**

  E-step: Update λ_1 given $\boldsymbol{\gamma}$
  $\mathbf{w}_t \leftarrow \big(w(t_1;\boldsymbol{\gamma}), \ldots, w(t_n;\boldsymbol{\gamma})\big)$
  $\lambda_1 \leftarrow \textsc{SolveLambda1}(\boldsymbol{\gamma}, \mathbf{t}, n)$ ▷ Same inner routine as Algorithm 1

  M-step: Update $\boldsymbol{\gamma}$ given λ_1
  function $\textsc{Objective}(\boldsymbol{\gamma}')$
    $\mathbf{w}'_t \leftarrow \big(w(t_1;\boldsymbol{\gamma}'), \ldots, w(t_n;\boldsymbol{\gamma}')\big)$
    $\mathbf{w}'_z \leftarrow \big(w(z_1;\boldsymbol{\gamma}'), \ldots, w(z_{n_1};\boldsymbol{\gamma}')\big)$
    if $\min(\mathbf{w}'_t) \le 0$ **then**
      return $-\infty$
    end if
    return $\sum_{j=1}^{n_1} \log w(z_j;\boldsymbol{\gamma}') - \sum_{i=1}^{n} \log\big(n + \lambda_1\big[w(t_i;\boldsymbol{\gamma}') - 1\big]\big)$
  end function
  $\boldsymbol{\gamma} \leftarrow \arg\max_{\boldsymbol{\gamma}'} \textsc{Objective}(\boldsymbol{\gamma}')$ ▷ BFGS with warm start $\boldsymbol{\gamma}$
  $\ell_{\text{new}} \leftarrow \textsc{Objective}(\boldsymbol{\gamma})$

  Convergence Check:
  if $|\ell_{\text{new}} - \ell_{\text{old}}| < \varepsilon \cdot (|\ell_{\text{old}}| + 1)$ **then**
    break
  end if
  $\ell_{\text{old}} \leftarrow \ell_{\text{new}}$
end for

Finalization:
$\hat{\boldsymbol{\gamma}} \leftarrow \boldsymbol{\gamma}$
$\mathbf{w}_t^{\text{final}} \leftarrow \big(w(t_1;\hat{\boldsymbol{\gamma}}), \ldots, w(t_n;\hat{\boldsymbol{\gamma}})\big)$
$\hat{\lambda}_1 \leftarrow \textsc{SolveLambda1}(\hat{\boldsymbol{\gamma}}, \mathbf{t}, n)$
$\hat{p}_i \leftarrow \big[n + \hat{\lambda}_1\big(w(t_i;\hat{\boldsymbol{\gamma}}) - 1\big)\big]^{-1}, \quad i = 1, \ldots, n$
return $(\hat{\boldsymbol{\gamma}}, \hat{\lambda}_1, \hat{p}_1, \ldots, \hat{p}_n)$

Computational Complexity. Each outer iteration performs one root-finding ($O(n)$) and one BFGS sub-optimization ($O(p^2 \cdot n)$). Convergence is linear but monotonic in the likelihood.
Remarks. Algorithm 3 is the most stable when initial values are poor or the likelihood surface is irregular. The E-step guarantees constraint satisfaction; the M-step guarantees likelihood increase. It is particularly suitable for complex weight functions where the joint Hessian in Algorithm 2 is expensive or ill-conditioned.

3.4. **Convergence Guarantees.** We now establish formal convergence guarantees for the proposed algorithms.

Theorem 3 (Convergence of Algorithm 1)**.** *Under the regularity conditions of Theorem 2, if $w(x;\boldsymbol{\gamma})$ is twice continuously differentiable and the profile log-likelihood (2.12) has a unique maximizer in the interior of Θ, then the sequence $\{\boldsymbol{\gamma}^{(k)}\}$ generated by Algorithm 1 converges to $\hat{\boldsymbol{\gamma}}$ for any starting value in a neighborhood of the optimum.*

Theorem 4 (Convergence of Algorithm 3)**.** *The sequence $\{(\boldsymbol{\gamma}^{(m)}, \lambda_1^{(m)})\}$ generated by Algorithm 3 satisfies $\ell_{\text{prof}}(\boldsymbol{\gamma}^{(m+1)}) \geq \ell_{\text{prof}}(\boldsymbol{\gamma}^{(m)})$ for all m. If the parameter space Θ is compact and $w(x;\boldsymbol{\gamma})$ is continuous, the sequence converges to a stationary point of ℓ_{prof}.*

Proof of Theorem 4. The E-step computes $\lambda_1^{(m)} = \lambda_1(\boldsymbol{\gamma}^{(m)})$ exactly, so the constraint is satisfied. The M-step maximizes the objective with λ_1 fixed, which is equivalent to maximizing ℓ_{prof} along a coordinate direction. Hence the likelihood is nondecreasing. By compactness and continuity, the sequence has a convergent subsequence; the limit must satisfy the first-order conditions. □

4. The Exponential Tilt Density Ratio Model

The exponential tilt density ratio model, originally introduced by Qin [1998], represents the most widely studied special case of the general DRM framework. In this section, we demonstrate how the general theory developed in Sections 2–3 specializes to this important case, and we highlight its connections to logistic regression and case-control studies.

4.1. **Model Specification.** The exponential tilt density ratio model assumes that the log-density ratio is linear in a known basis function $\mathbf{r}(x) = (r_1(x), \ldots, r_p(x))^\top$:

$$w(x;\alpha,\boldsymbol{\beta}) = \exp\left\{\alpha + \boldsymbol{\beta}^\top \mathbf{r}(x)\right\}, \tag{4.1}$$

where $\boldsymbol{\gamma} = (\alpha, \boldsymbol{\beta}^\top)^\top \in \mathbb{R}^{p+1}$. The parameter α serves as a normalizing constant ensuring $\int_{\mathscr{D}} w(x;\alpha,\boldsymbol{\beta})\, dG_0(x) = 1$, while $\boldsymbol{\beta}$ captures the distributional shift between populations.

Proposition 2 (Normalization of α)**.** *In the exponential tilt model (4.1), the parameter α is implicitly determined by $\boldsymbol{\beta}$ through*

$$\alpha(\boldsymbol{\beta}) = -\log \int_{\mathscr{D}} \exp\left\{\boldsymbol{\beta}^\top \mathbf{r}(x)\right\} dG_0(x). \tag{4.2}$$

Consequently, the effective parameter dimension is p rather than $p+1$.

Proof. The constraint $\int w(x;\alpha,\boldsymbol{\beta})\, dG_0(x) = 1$ yields

$$e^{\alpha} \int_{\mathscr{D}} e^{\boldsymbol{\beta}^\top \mathbf{r}(x)} dG_0(x) = 1,$$

which immediately gives (4.2). □

4.2. **Connection to Logistic Regression.** The exponential tilt model is intimately connected to prospective logistic regression under case-control sampling. Consider the prospective logistic model

$$\Pr(Y = 1 \mid X = x) = \frac{e^{\beta_0 + \boldsymbol{\beta}^\top \mathbf{r}(x)}}{1 + e^{\beta_0 + \boldsymbol{\beta}^\top \mathbf{r}(x)}}, \tag{4.3}$$

where $Y \in \{0, 1\}$ indicates case ($Y = 1$) or control ($Y = 0$) status. By Bayes' theorem, the retrospective density ratio satisfies

$$\frac{g_1(x)}{g_0(x)} = \frac{\Pr(Y = 1 \mid X = x)}{\Pr(Y = 0 \mid X = x)} \cdot \frac{\Pr(Y = 0)}{\Pr(Y = 1)} = \exp\left\{\alpha + \boldsymbol{\beta}^\top \mathbf{r}(x)\right\}, \tag{4.4}$$

where $\alpha = \beta_0 + \log[\Pr(Y = 0)/\Pr(Y = 1)]$. This confirms that the exponential tilt DRM is precisely the retrospective formulation of logistic regression.

Theorem 5 (Equivalence to Multinomial Regression)**.** *Under the exponential tilt model with combined sample* $\{t_1, \ldots, t_n\}$*, the profile log-likelihood* (2.12) *is equivalent to the log-likelihood of a multinomial logistic regression model with offset* $\log(n_1/n_0)$*. Specifically,*

$$\ell_{\text{prof}}(\boldsymbol{\beta}) = \sum_{i=1}^{n} \{\delta_i \log \pi_i + (1 - \delta_i) \log(1 - \pi_i)\} + \textit{const}, \tag{4.5}$$

where $\delta_i = \mathbf{1}_{\{t_i \in \textit{cases}\}}$ *and*

$$\pi_i = \frac{e^{\boldsymbol{\beta}^\top \mathbf{r}(t_i)}}{n_0/n_1 + e^{\boldsymbol{\beta}^\top \mathbf{r}(t_i)}}. \tag{4.6}$$

Proof. Substituting (4.1) into the profile log-likelihood (2.12) and eliminating α via the constraint yields

$$\ell_{\text{prof}}(\boldsymbol{\beta}) = \sum_{j=1}^{n_1} \boldsymbol{\beta}^\top \mathbf{r}(z_j) - \sum_{i=1}^{n} \log\left\{n_0 + n_1 e^{\boldsymbol{\beta}^\top \mathbf{r}(t_i)}\right\} + \text{const}.$$

Reparameterizing with $\pi_i = n_1 e^{\boldsymbol{\beta}^\top \mathbf{r}(t_i)}/(n_0 + n_1 e^{\boldsymbol{\beta}^\top \mathbf{r}(t_i)})$ gives the claimed multinomial form. □

Remark 6. Theorem 5 is of profound practical importance: it allows practitioners to fit the semi-parametric exponential tilt DRM using standard GLM software (e.g., `glm` in R) by treating the case-control indicators as responses and the basis functions $\mathbf{r}(x)$ as covariates, with case weights reflecting the sampling design.

4.3. **Specialized Algorithms.** The exponential tilt structure permits significant simplification of the algorithms in Sections 3.1–3.3.

Corollary 7 (Closed-Form λ_1 for Exponential Tilt)**.** *For the exponential tilt model, the implicit equation* (2.11) *for* λ_1 *admits the closed-form solution*

$$\lambda_1 = n_1 - n_0, \tag{4.7}$$

independent of $\boldsymbol{\beta}$*.*

Proof. Substituting $w(x; \alpha, \boldsymbol{\beta}) = e^{\alpha + \boldsymbol{\beta}^\top \mathbf{r}(x)}$ into (2.11) and using the normalization constraint yields

$$\sum_{i=1}^{n} \frac{e^{\alpha + \boldsymbol{\beta}^\top \mathbf{r}(t_i)} - 1}{n + \lambda_1 (e^{\alpha + \boldsymbol{\beta}^\top \mathbf{r}(t_i)} - 1)} = 0.$$

Multiplying through by the denominators and simplifying using $\sum_{i=1}^n p_i = 1$ and $\sum_{i=1}^n q_i = 1$ gives $\lambda_1 = n_1 - n_0$. □

The closed-form expression (4.7) eliminates the inner root-finding subroutine in Algorithm 1, reducing the computational complexity from $O(p^2 \cdot n \cdot N_{\text{root}})$ to $O(p^2 \cdot n)$. Algorithm 2 likewise simplifies, as the augmented system collapses to optimization over $\boldsymbol{\beta}$ alone.

4.4. **Asymptotic Distribution.** For the exponential tilt model, the profile information matrix takes an especially tractable form. Let

$$\bar{\mathbf{r}}(x;\boldsymbol{\beta}) = \mathbf{r}(x) - \frac{\sum_{i=1}^n \mathbf{r}(t_i) e^{\boldsymbol{\beta}^\top \mathbf{r}(t_i)}/d_i}{\sum_{i=1}^n e^{\boldsymbol{\beta}^\top \mathbf{r}(t_i)}/d_i}, \qquad d_i = n_0 + n_1 e^{\boldsymbol{\beta}^\top \mathbf{r}(t_i)}. \tag{4.8}$$

Corollary 8 (Asymptotic Normality for Exponential Tilt)**.** *Under the exponential tilt model with true parameter $\boldsymbol{\beta}_0$,*

$$\sqrt{n}(\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} \mathcal{N}_p\left(\mathbf{0}, \boldsymbol{\Sigma}_{\text{exp}}^{-1}\right), \tag{4.9}$$

where

$$\boldsymbol{\Sigma}_{\text{exp}} = \frac{n_0 n_1}{n^2}\,\mathbb{E}_0\left[\bar{\mathbf{r}}(X;\boldsymbol{\beta}_0)\bar{\mathbf{r}}(X;\boldsymbol{\beta}_0)^\top e^{\boldsymbol{\beta}_0^\top \mathbf{r}(X)}\right]. \tag{4.10}$$

Proof. This follows from Theorem 14 (to be established in Section 6) by direct computation of the profile information matrix, noting that the score with respect to $\boldsymbol{\beta}$ decomposes into the centered basis functions $\bar{\mathbf{r}}(x;\boldsymbol{\beta})$. □

5. Extension to Multiple Populations

We now extend the two-sample density ratio framework to accommodate $K \geq 2$ populations. This extension is essential for applications involving multiple treatment arms, multi-class classification, and stratified sampling designs with more than two strata.

5.1. **Model Formulation.** Let $G_0, G_1, \ldots, G_K$ denote the cumulative distribution functions of $K+1$ independent populations with densities $g_0, g_1, \ldots, g_K$, respectively. The K-sample density ratio model postulates

$$g_k(x) = w_k(x;\boldsymbol{\gamma}_k) g_0(x), \qquad k = 1, \ldots, K, \tag{5.1}$$

where each $w_k(x;\boldsymbol{\gamma}_k) > 0$ is a density ratio function with parameter vector $\boldsymbol{\gamma}_k \in \mathbb{R}^{p_k}$. The baseline density g_0 remains unspecified.

The constraints generalizing (2.5) are

$$\int_{\mathscr{D}} w_k(x;\boldsymbol{\gamma}_k)\, dG_0(x) = 1, \qquad k = 1, \ldots, K. \tag{5.2}$$

We observe independent random samples

$$x_{01}, \ldots, x_{0n_0} \overset{\text{i.i.d.}}{\sim} g_0(x), \qquad x_{k1}, \ldots, x_{kn_k} \overset{\text{i.i.d.}}{\sim} g_k(x), \quad k = 1, \ldots, K. \tag{5.3}$$

Let $n = \sum_{k=0}^K n_k$ denote the total sample size.

5.2. **Empirical Likelihood.** Let $\{t_1,\dots,t_n\}$ denote the pooled sample. Define $p_i = dG_0(t_i)$ and $q_{ki} = dG_k(t_i) = p_i w_k(t_i;\boldsymbol{\gamma}_k)$. The semiparametric empirical likelihood is

$$\mathscr{L}(\boldsymbol{\gamma},\mathbf{p}) = \prod_{i=1}^{n} p_i \prod_{k=1}^{K}\prod_{j=1}^{n_k} w_k(x_{kj};\boldsymbol{\gamma}_k), \tag{5.4}$$

subject to

$$\sum_{i=1}^{n} p_i = 1, \qquad \sum_{i=1}^{n} p_i\big[w_k(t_i;\boldsymbol{\gamma}_k)-1\big] = 0, \quad k=1,\dots,K. \tag{5.5}$$

Introducing Lagrange multipliers λ_0 for the normalization constraint and λ_k for the k-th DRM constraint, the Lagrangian becomes

$$\begin{aligned} Q(\boldsymbol{\gamma},\mathbf{p},\boldsymbol{\lambda}) = {} & \sum_{i=1}^{n}\log p_i + \sum_{k=1}^{K}\sum_{j=1}^{n_k}\log w_k(x_{kj};\boldsymbol{\gamma}_k) \\ & - \lambda_0\left(\sum_{i=1}^{n} p_i - 1\right) - \sum_{k=1}^{K}\lambda_k\sum_{i=1}^{n} p_i\big[w_k(t_i;\boldsymbol{\gamma}_k)-1\big]. \end{aligned} \tag{5.6}$$

Proposition 3 (MLE of Baseline Masses)**.** *The maximizing baseline probabilities are*

$$p_i = \frac{1}{n+\sum_{k=1}^{K}\lambda_k\big[w_k(t_i;\boldsymbol{\gamma}_k)-1\big]}, \qquad i=1,\dots,n, \tag{5.7}$$

where $\lambda_0 = n$ and $(\lambda_1,\dots,\lambda_K)$ satisfy the system of K implicit equations:

$$\Phi_k(\boldsymbol{\lambda};\boldsymbol{\gamma}) \equiv \sum_{i=1}^{n}\frac{w_k(t_i;\boldsymbol{\gamma}_k)-1}{n+\sum_{\ell=1}^{K}\lambda_\ell\big[w_\ell(t_i;\boldsymbol{\gamma}_\ell)-1\big]} = 0, \quad k=1,\dots,K. \tag{5.8}$$

Proof. The derivation mirrors that of Proposition 1. The derivative of (5.6) with respect to p_i yields

$$\frac{1}{p_i} - \lambda_0 - \sum_{k=1}^{K}\lambda_k\big[w_k(t_i;\boldsymbol{\gamma}_k)-1\big] = 0.$$

Solving for p_i and applying the normalization constraint gives $\lambda_0 = n$ and (5.7). Substituting into the DRM constraints yields (5.8). □

Theorem 9 (Existence and Uniqueness)**.** *For fixed $\boldsymbol{\gamma} = (\boldsymbol{\gamma}_1^\top,\dots,\boldsymbol{\gamma}_K^\top)^\top$, if the matrix with entries*

$$M_{k\ell} = \sum_{i=1}^{n}\frac{\big[w_k(t_i;\boldsymbol{\gamma}_k)-1\big]\big[w_\ell(t_i;\boldsymbol{\gamma}_\ell)-1\big]}{\big\{n+\sum_{m=1}^{K}\lambda_m\big[w_m(t_i;\boldsymbol{\gamma}_m)-1\big]\big\}^2} \tag{5.9}$$

is positive definite for all $\boldsymbol{\lambda}$ in the feasible region, then the system (5.8) has a unique solution $\boldsymbol{\lambda}(\boldsymbol{\gamma}) = (\lambda_1(\boldsymbol{\gamma}),\dots,\lambda_K(\boldsymbol{\gamma}))^\top$.

Proof. The Jacobian of the system $\boldsymbol{\Phi} = (\Phi_1,\dots,\Phi_K)^\top$ with respect to $\boldsymbol{\lambda}$ is precisely $-\mathbf{M}$. Positive definiteness of $\mathbf{M}$ implies that $\boldsymbol{\Phi}$ is strongly monotone, guaranteeing a unique root by standard results in variational inequalities. See more details in Facchinei and Pang [2003]. □

5.3. **Profile Log-Likelihood and Estimation.** Substituting (5.7) into (5.4) yields the profile log-likelihood

$$\ell_{\text{prof}}(\boldsymbol{\gamma}) = \sum_{k=1}^{K}\sum_{j=1}^{n_k}\log w_k(x_{kj};\boldsymbol{\gamma}_k) - \sum_{i=1}^{n}\log\left\{n + \sum_{\ell=1}^{K}\lambda_\ell(\boldsymbol{\gamma})\big[w_\ell(t_i;\boldsymbol{\gamma}_\ell) - 1\big]\right\} - n\log n. \tag{5.10}$$

The gradient of the profile log-likelihood generalizes (2.13) through the implicit function theorem applied to the system (5.8). Let $\boldsymbol{\gamma} = (\boldsymbol{\gamma}_1^\top, \ldots, \boldsymbol{\gamma}_K^\top)^\top \in \mathbb{R}^P$ where $P = \sum_{k=1}^{K} p_k$. The score function is

$$\mathbf{S}_{\text{prof}}(\boldsymbol{\gamma}) = \begin{pmatrix} \displaystyle\sum_{j=1}^{n_1}\frac{\nabla_{\boldsymbol{\gamma}_1}w_1(x_{1j};\boldsymbol{\gamma}_1)}{w_1(x_{1j};\boldsymbol{\gamma}_1)} - \sum_{i=1}^{n}\frac{\lambda_1(\boldsymbol{\gamma})\nabla_{\boldsymbol{\gamma}_1}w_1(t_i;\boldsymbol{\gamma}_1)}{d_i(\boldsymbol{\gamma})} \\ \vdots \\ \displaystyle\sum_{j=1}^{n_K}\frac{\nabla_{\boldsymbol{\gamma}_K}w_K(x_{Kj};\boldsymbol{\gamma}_K)}{w_K(x_{Kj};\boldsymbol{\gamma}_K)} - \sum_{i=1}^{n}\frac{\lambda_K(\boldsymbol{\gamma})\nabla_{\boldsymbol{\gamma}_K}w_K(t_i;\boldsymbol{\gamma}_K)}{d_i(\boldsymbol{\gamma})} \end{pmatrix}, \tag{5.11}$$

where $d_i(\boldsymbol{\gamma}) = n + \sum_{\ell=1}^{K}\lambda_\ell(\boldsymbol{\gamma})\big[w_\ell(t_i;\boldsymbol{\gamma}_\ell) - 1\big]$.

The gradient of $\boldsymbol{\lambda}(\boldsymbol{\gamma})$ is obtained by solving the linear system

$$\mathbf{M}(\boldsymbol{\gamma})\nabla_{\boldsymbol{\gamma}}\boldsymbol{\lambda}(\boldsymbol{\gamma}) = -\mathbf{N}(\boldsymbol{\gamma}), \tag{5.12}$$

where $\mathbf{M}(\boldsymbol{\gamma})$ is as in Theorem 9 and $\mathbf{N}(\boldsymbol{\gamma})$ is the $K \times P$ matrix with blocks

$$N_{k,\boldsymbol{\gamma}_\ell} = \sum_{i=1}^{n}\frac{n\nabla_{\boldsymbol{\gamma}_\ell}w_\ell(t_i;\boldsymbol{\gamma}_\ell)\mathbf{1}_{\{k=\ell\}}\big[w_k(t_i;\boldsymbol{\gamma}_k) - 1\big] - n\lambda_k(\boldsymbol{\gamma})\nabla_{\boldsymbol{\gamma}_\ell}w_\ell(t_i;\boldsymbol{\gamma}_\ell)}{d_i(\boldsymbol{\gamma})^2}. \tag{5.13}$$

5.4. **Algorithms for Multiple Populations.** The algorithms of Sections 3.1–3.3 extend naturally to the multi-population setting.

Algorithm 1′ (Nested Optimization): The outer loop optimizes over $\boldsymbol{\gamma} \in \mathbb{R}^P$ while the inner loop solves the K-dimensional system (5.8) via Newton's method or Broyden's method. The computational complexity becomes $O(P^2 \cdot n \cdot N_{\text{root}}^{(K)})$, where $N_{\text{root}}^{(K)}$ is the average number of iterations for the K-dimensional root-finding.

Algorithm 2′ (Joint Newton–Raphson): The augmented parameter is $\boldsymbol{\theta} = (\boldsymbol{\gamma}^\top, \boldsymbol{\lambda}^\top)^\top \in \mathbb{R}^{P+K}$. The score system $\mathbf{S}(\boldsymbol{\theta}) = (\mathbf{S}_{\text{prof}}(\boldsymbol{\gamma})^\top, \boldsymbol{\Phi}(\boldsymbol{\lambda};\boldsymbol{\gamma})^\top)^\top = \mathbf{0}$ is solved via Newton–Raphson with a $(P+K)\times(P+K)$ Hessian. Each iteration costs $O((P+K)^2 \cdot n + (P+K)^3)$.

Algorithm 3′ (EM-Type): The E-step updates all K multipliers $\boldsymbol{\lambda}$ given $\boldsymbol{\gamma}$; the M-step optimizes $\boldsymbol{\gamma}$ given $\boldsymbol{\lambda}$. The convergence properties of Theorem 4 carry over directly.

5.5. **Asymptotic Theory.** Let $\boldsymbol{\gamma}_0 = (\boldsymbol{\gamma}_{1,0}^\top, \ldots, \boldsymbol{\gamma}_{K,0}^\top)^\top$ denote the true parameter. Define the block-diagonal information matrix

$$\mathscr{I}_{\text{prof}}^{\text{multi}}(\boldsymbol{\gamma}) = \operatorname{diag}\big\{I_{11}^{(1)}(\boldsymbol{\gamma}), \ldots, I_{11}^{(K)}(\boldsymbol{\gamma})\big\} - \mathbf{I}_{12}(\boldsymbol{\gamma})\mathbf{I}_{22}(\boldsymbol{\gamma})^{-1}\mathbf{I}_{12}(\boldsymbol{\gamma})^\top, \tag{5.14}$$

where $I_{11}^{(k)}$ is the information for $\boldsymbol{\gamma}_k$ from the k-th sample, $\mathbf{I}_{12}$ is the $P \times K$ cross-information matrix, and $\mathbf{I}_{22}$ is the $K \times K$ multiplier information matrix generalizing (6.3).

Theorem 10 (Asymptotic Normality in Multi-Sample DRM). *Under regularity conditions extending Assumptions 7.1–7.1 to $K+1$ populations, with $n_k/n \to \rho_k \in (0,1)$ for $k = 0, \ldots, K$,*

$$\sqrt{n}(\tilde{\boldsymbol{\gamma}} - \boldsymbol{\gamma}_0) \xrightarrow{d} \mathscr{N}_P\left(\mathbf{0}, \mathscr{I}_{\text{prof}}^{\text{multi}}(\boldsymbol{\gamma}_0)^{-1}\right). \tag{5.15}$$

Theorem 11 (Weak Convergence of Multi-Sample CDFs). *The estimated cumulative distribution functions*

$$\tilde{G}_k(x) = \sum_{i:t_i \le x} \frac{w_k(t_i; \tilde{\boldsymbol{\gamma}}_k)}{n + \sum_{\ell=1}^K \tilde{\lambda}_\ell \left[w_\ell(t_i; \tilde{\boldsymbol{\gamma}}_\ell) - 1\right]}, \qquad k = 0, \ldots, K, \tag{5.16}$$

satisfy joint weak convergence to a tight Gaussian process. The limiting covariance structure generalizes Theorem 16 with cross-covariance terms reflecting the shared dependence on the pooled sample.

5.6. **Special Case: Exponential Tilt for Multiple Populations.** When all ratio functions share a common exponential tilt structure,

$$w_k(x; \boldsymbol{\beta}_k) = \exp\left\{\alpha_k + \boldsymbol{\beta}_k^\top \mathbf{r}(x)\right\}, \qquad k = 1, \ldots, K, \tag{5.17}$$

the model reduces to multinomial logistic regression with $K+1$ classes. The normalizing constants satisfy $\alpha_k = -\log \int e^{\boldsymbol{\beta}_k^\top \mathbf{r}(x)} dG_0(x)$, and the profile log-likelihood becomes

$$\ell_{\text{prof}}(\boldsymbol{\beta}_1, \ldots, \boldsymbol{\beta}_K) = \sum_{k=1}^K \sum_{j=1}^{n_k} \boldsymbol{\beta}_k^\top \mathbf{r}(x_{kj}) - \sum_{i=1}^n \log\left\{\sum_{\ell=0}^K \frac{n_\ell}{n} e^{\boldsymbol{\beta}_\ell^\top \mathbf{r}(t_i)}\right\} + \text{const}, \tag{5.18}$$

with $\boldsymbol{\beta}_0 \equiv \mathbf{0}$ for identifiability. This is precisely the likelihood of a multinomial logit model with offsets $\log(n_k/n_0)$.

6. Asymptotic Theory for Maximum Likelihood Estimation

In this section, we establish the asymptotic properties of the maximum empirical likelihood estimators derived in Sections 2–3. We begin with consistency and asymptotic normality of the parameter estimator $\tilde{\boldsymbol{\gamma}}$, followed by weak convergence of the estimated cumulative distribution functions $\tilde{G}_0$ and $\tilde{G}_1$.

6.1. **Regularity Conditions.** We impose the following regularity conditions on the density ratio function $w(x; \boldsymbol{\gamma})$ and the underlying distributions.

Assumption 12 (Smoothness and Identifiability). The density ratio function $w(x; \boldsymbol{\gamma})$ satisfies:

- (A1) $w(x; \boldsymbol{\gamma}) > 0$ for all $x \in \mathscr{D}$ and $\boldsymbol{\gamma} \in \Theta$, where $\Theta \subset \mathbb{R}^p$ is a compact parameter space;
- (A2) $w(x; \boldsymbol{\gamma})$ is twice continuously differentiable with respect to $\boldsymbol{\gamma}$ for each $x \in \mathscr{D}$;
- (A3) The matrix $\mathbb{E}_0\left[\nabla_{\boldsymbol{\gamma}} w(X; \boldsymbol{\gamma}_0) \nabla_{\boldsymbol{\gamma}} w(X; \boldsymbol{\gamma}_0)^\top / w(X; \boldsymbol{\gamma}_0)^2\right]$ is positive definite, where $\mathbb{E}_0$ denotes expectation under G_0;
- (A4) There exists a neighborhood $\mathscr{N}$ of the true parameter $\boldsymbol{\gamma}_0$ such that

$$\mathbb{E}_0\left[\sup_{\boldsymbol{\gamma} \in \mathscr{N}} \|\nabla_{\boldsymbol{\gamma}} w(X; \boldsymbol{\gamma})\|^2\right] < \infty, \quad \mathbb{E}_0\left[\sup_{\boldsymbol{\gamma} \in \mathscr{N}} \|\nabla^2_{\boldsymbol{\gamma}\boldsymbol{\gamma}} w(X; \boldsymbol{\gamma})\|\right] < \infty.$$

Assumption 13 (Sampling Design). The sample sizes satisfy $n_0, n_1 \to \infty$ with $n_0/n \to \rho \in (0,1)$, where $n = n_0 + n_1$.

6.2. **Asymptotic Normality of $\tilde{\gamma}$.** Let γ_0 denote the true parameter value. Define the following information quantities:

$$I_{11}(\boldsymbol{\gamma}) = \frac{1}{n}\sum_{i=1}^{n} \frac{\nabla_{\boldsymbol{\gamma}} w(t_i;\boldsymbol{\gamma})\nabla_{\boldsymbol{\gamma}} w(t_i;\boldsymbol{\gamma})^\top}{\{n+\lambda_1(\boldsymbol{\gamma})[w(t_i;\boldsymbol{\gamma})-1]\}^2}, \tag{6.1}$$

$$I_{12}(\boldsymbol{\gamma}) = \frac{1}{n}\sum_{i=1}^{n} \frac{\nabla_{\boldsymbol{\gamma}} w(t_i;\boldsymbol{\gamma})\big[w(t_i;\boldsymbol{\gamma})-1\big]}{\{n+\lambda_1(\boldsymbol{\gamma})[w(t_i;\boldsymbol{\gamma})-1]\}^2}, \tag{6.2}$$

$$I_{22}(\boldsymbol{\gamma}) = \frac{1}{n}\sum_{i=1}^{n} \frac{\big[w(t_i;\boldsymbol{\gamma})-1\big]^2}{\{n+\lambda_1(\boldsymbol{\gamma})[w(t_i;\boldsymbol{\gamma})-1]\}^2}. \tag{6.3}$$

The profile information matrix for $\boldsymbol{\gamma}$ is given by the Schur complement:

$$\mathscr{I}_{\text{prof}}(\boldsymbol{\gamma}) = I_{11}(\boldsymbol{\gamma}) - \frac{I_{12}(\boldsymbol{\gamma})I_{12}(\boldsymbol{\gamma})^\top}{I_{22}(\boldsymbol{\gamma})}. \tag{6.4}$$

Theorem 14 (Asymptotic Normality of $\tilde{\gamma}$)**.** *Under Assumptions 7.1 and 7.1, the maximum empirical likelihood estimator $\tilde{\gamma}$ satisfies*

$$\sqrt{n}(\tilde{\boldsymbol{\gamma}} - \boldsymbol{\gamma}_0) \xrightarrow{d} \mathscr{N}_p\left(\mathbf{0}, \mathscr{I}_{\text{prof}}(\boldsymbol{\gamma}_0)^{-1}\right), \tag{6.5}$$

where $\mathscr{I}_{\text{prof}}(\boldsymbol{\gamma}_0)$ is the limiting profile information matrix defined in (6.4). Moreover, the asymptotic variance can be consistently estimated by $\mathscr{I}_{\text{prof}}(\tilde{\boldsymbol{\gamma}})^{-1}/n$.

Proof. The proof proceeds in three steps.

Step 1: Score decomposition. The profile score function is

$$\mathbf{S}_{\text{prof}}(\boldsymbol{\gamma}) = \sum_{j=1}^{n_1} \frac{\nabla_{\boldsymbol{\gamma}} w(z_j;\boldsymbol{\gamma})}{w(z_j;\boldsymbol{\gamma})} - \sum_{i=1}^{n} \frac{\lambda_1(\boldsymbol{\gamma})\nabla_{\boldsymbol{\gamma}} w(t_i;\boldsymbol{\gamma}) + \big[w(t_i;\boldsymbol{\gamma})-1\big]\nabla_{\boldsymbol{\gamma}}\lambda_1(\boldsymbol{\gamma})}{n+\lambda_1(\boldsymbol{\gamma})\big[w(t_i;\boldsymbol{\gamma})-1\big]}.$$

By the implicit function theorem applied to the constraint equation (2.11), we have

$$\nabla_{\boldsymbol{\gamma}}\lambda_1(\boldsymbol{\gamma}) = -\frac{\sum_{i=1}^{n} \frac{n\nabla_{\boldsymbol{\gamma}} w(t_i;\boldsymbol{\gamma})}{\{n+\lambda_1[w(t_i;\boldsymbol{\gamma})-1]\}^2}}{\sum_{i=1}^{n} \frac{[w(t_i;\boldsymbol{\gamma})-1]^2}{\{n+\lambda_1[w(t_i;\boldsymbol{\gamma})-1]\}^2}} = -\frac{I_{12}(\boldsymbol{\gamma})}{I_{22}(\boldsymbol{\gamma})}.$$

Step 2: Linearization. Expanding $\mathbf{S}_{\text{prof}}(\tilde{\boldsymbol{\gamma}}) = \mathbf{0}$ around $\boldsymbol{\gamma}_0$ yields

$$\mathbf{0} = \mathbf{S}_{\text{prof}}(\boldsymbol{\gamma}_0) + \nabla_{\boldsymbol{\gamma}}\mathbf{S}_{\text{prof}}(\boldsymbol{\gamma}_0)(\tilde{\boldsymbol{\gamma}} - \boldsymbol{\gamma}_0) + o_p(\sqrt{n}\|\tilde{\boldsymbol{\gamma}} - \boldsymbol{\gamma}_0\|).$$

By the law of large numbers and the continuous mapping theorem,

$$-\frac{1}{n}\nabla_{\boldsymbol{\gamma}}\mathbf{S}_{\text{prof}}(\boldsymbol{\gamma}_0) \xrightarrow{p} \mathscr{I}_{\text{prof}}(\boldsymbol{\gamma}_0).$$

Step 3: Central limit theorem. The normalized score satisfies

$$\frac{1}{\sqrt{n}}\mathbf{S}_{\text{prof}}(\boldsymbol{\gamma}_0) \xrightarrow{d} \mathscr{N}_p\big(\mathbf{0}, \mathscr{I}_{\text{prof}}(\boldsymbol{\gamma}_0)\big).$$

This follows from the martingale central limit theorem applied to the empirical likelihood score, noting that the two samples are independent and the constraint structure induces the appropriate variance stabilization. Combining Steps 2 and 3 via Slutsky's theorem yields the result. □

Corollary 15 (Wald-type Confidence Regions)**.** *Under the conditions of Theorem 14, for any* $\alpha \in (0,1)$*,*

$$n(\tilde{\boldsymbol{\gamma}} - \boldsymbol{\gamma}_0)^\top \mathscr{I}_{\text{prof}}(\tilde{\boldsymbol{\gamma}})(\tilde{\boldsymbol{\gamma}} - \boldsymbol{\gamma}_0) \xrightarrow{d} \chi^2_p, \tag{6.6}$$

so that an asymptotic $100(1-\alpha)\%$ *confidence ellipsoid for* $\boldsymbol{\gamma}_0$ *is*

$$\left\{\boldsymbol{\gamma} : n(\boldsymbol{\gamma} - \tilde{\boldsymbol{\gamma}})^\top \mathscr{I}_{\text{prof}}(\tilde{\boldsymbol{\gamma}})(\boldsymbol{\gamma} - \tilde{\boldsymbol{\gamma}}) \le \chi^2_{p,1-\alpha}\right\}. \tag{6.7}$$

6.3. **Weak Convergence of $\tilde{G}_0$ and $\tilde{G}_1$.** The maximum empirical likelihood estimators of the baseline and tilted cumulative distribution functions are given by

$$\tilde{G}_0(x) = \sum_{i:t_i \le x} \tilde{p}_i = \sum_{i:t_i\le x} \frac{1}{n + \tilde{\lambda}_1\big[w(t_i;\tilde{\boldsymbol{\gamma}}) - 1\big]}, \tag{6.8}$$

$$\tilde{G}_1(x) = \sum_{i:t_i \le x} \tilde{q}_i = \sum_{i:t_i\le x} \frac{w(t_i;\tilde{\boldsymbol{\gamma}})}{n + \tilde{\lambda}_1\big[w(t_i;\tilde{\boldsymbol{\gamma}}) - 1\big]}. \tag{6.9}$$

Let G_0^0 and G_1^0 denote the true cumulative distribution functions. Define the empirical processes

$$\mathbb{G}_{n,0}(x) = \sqrt{n}\big\{\tilde{G}_0(x) - G_0^0(x)\big\}, \quad \mathbb{G}_{n,1}(x) = \sqrt{n}\big\{\tilde{G}_1(x) - G_1^0(x)\big\}. \tag{6.10}$$

Theorem 16 (Weak Convergence of $\tilde{G}_0$ and $\tilde{G}_1$)**.** *Under Assumptions 7.1 and 7.1, the processes* $(\mathbb{G}_{n,0}, \mathbb{G}_{n,1})$ *converge weakly in* $\ell^\infty(\mathbb{R}) \times \ell^\infty(\mathbb{R})$ *to a tight Gaussian process* $(\mathbb{G}_0, \mathbb{G}_1)$ *with mean zero and covariance structure characterized as follows.*

Let $H_0(x) = \mathbb{E}_0\big[\mathbf{1}_{\{X\le x\}} \nabla_{\boldsymbol{\gamma}} w(X;\boldsymbol{\gamma}_0)/w(X;\boldsymbol{\gamma}_0)\big]$ *and define*

$$\Sigma_0(x,y) = G_0^0(x\wedge y) - G_0^0(x)G_0^0(y) - H_0(x)^\top \mathscr{I}_{\text{prof}}(\boldsymbol{\gamma}_0)^{-1} H_0(y). \tag{6.11}$$

Then for all $x, y \in \mathbb{R}$*:*

$$\mathrm{Cov}\big(\mathbb{G}_0(x), \mathbb{G}_0(y)\big) = \frac{1}{\rho}\Sigma_0(x,y), \tag{6.12}$$

$$\mathrm{Cov}\big(\mathbb{G}_1(x), \mathbb{G}_1(y)\big) = \frac{1}{1-\rho}\Sigma_1(x,y) + \frac{1}{\rho} H_1(x)^\top \mathscr{I}_{\text{prof}}(\boldsymbol{\gamma}_0)^{-1} H_1(y), \tag{6.13}$$

$$\mathrm{Cov}\big(\mathbb{G}_0(x), \mathbb{G}_1(y)\big) = -\frac{1}{\rho} H_0(x)^\top \mathscr{I}_{\text{prof}}(\boldsymbol{\gamma}_0)^{-1} H_1(y), \tag{6.14}$$

where Σ_1 *and* H_1 *are the corresponding quantities under* G_1^0*.*

Proof. The proof relies on the functional delta method and the asymptotic linear representation of the estimators.

Step 1: Asymptotic linearity. We first establish that

$$\mathbb{G}_{n,0}(x) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_{0,x}(t_i) + o_p(1), \tag{6.15}$$

where the influence function is

$$\psi_{0,x}(t) = \mathbf{1}_{\{t\le x\}} - G_0^0(x) - H_0(x)^\top \mathscr{I}_{\text{prof}}(\boldsymbol{\gamma}_0)^{-1} \left\{ \frac{n_1}{n} \frac{\nabla_{\boldsymbol{\gamma}} w(t;\boldsymbol{\gamma}_0)}{w(t;\boldsymbol{\gamma}_0)} - \frac{\lambda_1^0 \nabla_{\boldsymbol{\gamma}} w(t;\boldsymbol{\gamma}_0)}{n + \lambda_1^0\big[w(t;\boldsymbol{\gamma}_0) - 1\big]} \right\}.$$

This representation follows from expanding (6.8) around $(\boldsymbol{\gamma}_0, \lambda_1^0)$ using the delta method, where $\lambda_1^0 = \lim_{n\to\infty} \lambda_1(\boldsymbol{\gamma}_0) = n_1 - n_0$ under the DRM constraint.

Step 2: Tightness. The empirical process $\{\mathbb{G}_{n,0}(x) : x \in \mathbb{R}\}$ is tight because the indicator functions $\mathbf{1}_{\{t_i \le x\}}$ form a VC-class, and the correction term involving $\tilde{\boldsymbol{\gamma}}$ is uniformly continuous by the smoothness assumptions on $w(x;\boldsymbol{\gamma})$.

Step 3: Finite-dimensional convergence. For any finite collection $x_1, \ldots, x_k$, the multivariate central limit theorem applies to $(\mathbb{G}_{n,0}(x_1), \ldots, \mathbb{G}_{n,0}(x_k))$ by the asymptotic linear representation (6.15). The covariance calculation yields (6.12) after accounting for the correlation induced by the estimated parameter $\tilde{\boldsymbol{\gamma}}$.

Step 4: Joint convergence. The joint weak convergence of $(\mathbb{G}_{n,0}, \mathbb{G}_{n,1})$ follows from the Cramér–Wold device and the observation that both processes depend on the same estimated parameter $\tilde{\boldsymbol{\gamma}}$. The cross-covariance (6.14) arises from this shared dependence. □

Remark 17. The covariance structure reveals an important efficiency property: the term

$$-H_0(x)^\top \mathscr{I}_{\text{prof}}(\boldsymbol{\gamma}_0)^{-1} H_0(y)$$

in (6.12) represents the efficiency gain from semiparametric estimation relative to the empirical distribution function based solely on the control sample. The profile information matrix $\mathscr{I}_{\text{prof}}(\boldsymbol{\gamma}_0)$ captures the information contributed by both samples through the density ratio constraint.

Corollary 18 (Asymptotic Distribution at a Fixed Point). *For any fixed $x \in \mathbb{R}$ where $g_0^0(x) > 0$,*

$$\sqrt{n}\begin{pmatrix} \tilde{G}_0(x) - G_0^0(x) \\ \tilde{G}_1(x) - G_1^0(x) \end{pmatrix} \xrightarrow{d} \mathscr{N}_2\left(\mathbf{0}, \begin{pmatrix} \sigma_0^2(x) & \sigma_{01}(x) \\ \sigma_{01}(x) & \sigma_1^2(x) \end{pmatrix}\right), \tag{6.16}$$

where $\sigma_0^2(x) = \operatorname{Cov}\big(\mathbb{G}_0(x), \mathbb{G}_0(x)\big)/\rho$, $\sigma_1^2(x) = \operatorname{Cov}\big(\mathbb{G}_1(x), \mathbb{G}_1(x)\big)/(1-\rho)$, *and* $\sigma_{01}(x) = \operatorname{Cov}\big(\mathbb{G}_0(x), \mathbb{G}_1(x)\big)/\sqrt{\rho(1-\rho)}$.

6.4. **Efficiency Considerations.** The semiparametric MLE $\tilde{G}_0$ achieves the semiparametric efficiency bound for estimating G_0 in the density ratio model. This can be seen by comparing the asymptotic variance in Theorem 16 with the variance of the nonparametric MLE based only on the control sample, which would be $G_0^0(x)(1 - G_0^0(x))/\rho$. The efficiency gain is strictly positive whenever $H_0(x) \neq \mathbf{0}$, which occurs when the density ratio $w(x;\boldsymbol{\gamma})$ genuinely depends on $\boldsymbol{\gamma}$.

Theorem 19 (Semiparametric Efficiency). *The estimator $\tilde{G}_0(x)$ is asymptotically efficient among all regular estimators of $G_0^0(x)$ in the semiparametric model defined by* (2.1)*. That is, for any regular estimator sequence $\{\hat{G}_{n,0}(x)\}$,*

$$\liminf_{n\to\infty} n \operatorname{Var}\big(\hat{G}_{n,0}(x)\big) \ge \frac{1}{\rho}\big\{G_0^0(x)(1 - G_0^0(x)) - H_0(x)^\top \mathscr{I}_{\text{prof}}(\boldsymbol{\gamma}_0)^{-1} H_0(x)\big\}. \tag{6.17}$$

Sketch of Proof. The tangent space for the density ratio model consists of score functions of the form $a(x) + \mathbf{b}^\top \nabla_{\boldsymbol{\gamma}} \log w(x;\boldsymbol{\gamma}_0)$, where $a(x)$ is mean-zero under G_0^0 and $\mathbf{b} \in \mathbb{R}^p$. The efficient influence function for estimating $G_0^0(x)$ is the projection of the nonparametric influence function $\mathbf{1}_{\{X \le x\}} - G_0^0(x)$ onto this tangent space. Direct calculation shows that this projection yields exactly the influence function $\psi_{0,x}$ derived in (6.15), confirming semiparametric efficiency. □

7. Semiparametric Efficiency

In this section we characterize the semiparametric efficiency bound for the GLM–based density ratio model

$$\frac{dG_1}{dG_0}(x) = \theta\, w(x;\boldsymbol{\gamma}), \qquad \boldsymbol{\gamma} \in \Theta \subset \mathbb{R}^{p+1}, \tag{7.1}$$

where $\theta > 0$ is a known design constant arising from the stratified sampling fraction. By reparametrizing the parameter vector so that $\alpha^* = \alpha + \log\theta$, the model becomes $dG_1/dG_0 = w(x;\boldsymbol{\gamma}^*)$ with the same structural form. Hence, without loss of generality we present the derivation for the canonical density ratio

$$\mathscr{P} = \Big\{(G_0, \boldsymbol{\gamma}) : \; g_1(x) = w(x;\boldsymbol{\gamma}) g_0(x), \; \int_{\mathscr{D}} w(x;\boldsymbol{\gamma})\, dG_0(x) = 1\Big\}, \tag{7.2}$$

where $w(x;\boldsymbol{\gamma}) > 0$ is a known positive function, smooth in $\boldsymbol{\gamma}$, and G_0 is an infinite–dimensional nuisance parameter. We write

$$\dot{\ell}_{\boldsymbol{\gamma}}(x) \coloneqq \nabla_{\boldsymbol{\gamma}} \log w(x;\boldsymbol{\gamma})$$

for the parametric score contributed by a single observation through the density ratio.

7.1. **Regularity conditions.** The following regularity conditions are assumed throughout this section. They ensure the validity of the semiparametric efficiency calculations and the asymptotic normality of the maximum empirical likelihood estimator for a general DRM $w(x;\boldsymbol{\gamma})$.

- C1. **Smoothness and integrability.** The density ratio $w(x;\boldsymbol{\gamma})$ is twice continuously differentiable with respect to $\boldsymbol{\gamma}$ on an open set $\Theta \subset \mathbb{R}^{p+1}$. Moreover,
$$\mathbb{E}_{G_0}\Big[\sup_{\boldsymbol{\gamma}\in\mathscr{N}} \big\|\nabla_{\boldsymbol{\gamma}} w(X;\boldsymbol{\gamma})\big\|^2\Big] < \infty \quad \text{and} \quad \mathbb{E}_{G_0}\Big[\sup_{\boldsymbol{\gamma}\in\mathscr{N}} \big\|\nabla^2_{\boldsymbol{\gamma}} w(X;\boldsymbol{\gamma})\big\|\Big] < \infty$$
for some neighborhood $\mathscr{N}$ of $\boldsymbol{\gamma}_0$. This guarantees that the parametric score $\dot{\ell}_{\boldsymbol{\gamma}}(x) = \nabla_{\boldsymbol{\gamma}} \log w(x;\boldsymbol{\gamma})$ and its derivative are well behaved in a neighborhood of the true parameter.
- C2. **Sampling design.** The sample sizes satisfy $n_0/n \to \rho \in (0,1)$ and $n_1/n \to 1-\rho$ as $n \to \infty$, where $n = n_0 + n_1$. The observations are independent within each stratum, and the limiting design ratio ρ is bounded away from 0 and 1.
- C3. **Identifiability.** The parametric score vector $\dot{\ell}_{\boldsymbol{\gamma}}(x)$ is not almost surely concentrated on a proper linear subspace of $\mathbb{R}^{p+1}$ under G_0; equivalently, the semiparametric information matrix $\mathscr{I}^*(\boldsymbol{\gamma}_0)$ defined in (7.11) is positive definite. This ensures that $\boldsymbol{\gamma}_0$ is locally identifiable and that the efficient score has full rank.
- C4. **Boundedness of the density ratio.** There exist constants $0 < c_1 \le c_2 < \infty$ such that $c_1 \le w(x;\boldsymbol{\gamma}) \le c_2$ for all $x \in \mathscr{D}$ and all $\boldsymbol{\gamma}$ in a neighborhood of $\boldsymbol{\gamma}_0$. This guarantees that the retrospective probability $\pi(x;\boldsymbol{\gamma})$ defined in (7.7) is bounded away from 0 and 1, ensuring a well–posed and invertible information matrix.
- C5. **Donsker class.** The class of functions $\mathscr{F} = \{x \mapsto \dot{\ell}_{\boldsymbol{\gamma}}(x) : \boldsymbol{\gamma} \in \Theta\}$ is G_0–Donsker with a square–integrable envelope function. This condition is needed for the uniform convergence of the empirical process associated with the profile score and information; see van der Vaart and Wellner [1996] for the general theory.

7.2. **Nuisance tangent space.** Let G_0 be the infinite–dimensional nuisance parameter. Consider a parametric submodel $G_0(\cdot;\eta)$ with $G_0(\cdot;0) = G_0$ and score function

$$a(x) = \frac{\partial}{\partial\eta} \log g_0(x;\eta)\Big|_{\eta=0},$$

satisfying $a \in L_2^0(G_0)$, i.e. $\mathbb{E}_{G_0}[a(X)] = 0$. Differentiating the normalization constraint $\int w(x;\boldsymbol{\gamma}) g_0(x;\eta)\, dx = 1$ with respect to η yields the additional restriction

$$\mathbb{E}_{G_0}\big[w(X;\boldsymbol{\gamma})\, a(X)\big] = 0. \tag{7.3}$$

Thus the nuisance tangent space for a single observation drawn from G_0 is

$$\mathscr{T} = \Big\{ a(X) \in L_2^0(G_0) :\ \mathbb{E}_{G_0}\big[w(X;\boldsymbol{\gamma})\, a(X)\big] = 0 \Big\}. \tag{7.4}$$

Geometrically, $\mathscr{T}$ is the orthogonal complement of $\operatorname{span}\{w(X;\boldsymbol{\gamma}) - 1\}$ in $L_2^0(G_0)$.

Under the stratified sampling design, the observed data are (X_i, δ_i), where $\delta_i = 1$ if X_i is a case (drawn from G_1) and $\delta_i = 0$ if it is a control (drawn from G_0). The density of (X, δ) with respect to $\mu = G_0 \times \text{counting}$ is

$$f(x, \delta) = \Big(\frac{n_0}{n}\Big)^{1-\delta} \Big(\frac{n_1}{n}\Big)^{\delta} \big[w(x;\boldsymbol{\gamma})\big]^{\delta} g_0(x).$$

Because $\partial_\eta \log f(x, \delta; \eta) = a(x)$ for both $\delta = 0$ and $\delta = 1$, the nuisance tangent space in the full Hilbert space $L_2^0(P)$ of mean–zero functions of (X, δ) is precisely the set of functions $a(X)$ (independent of δ) satisfying (7.3).

7.3. **Parametric score and its projection.** The parametric score for $\boldsymbol{\gamma}$ contributed by a case observation is

$$\dot{\ell}_{\boldsymbol{\gamma}}(X) = \nabla_{\boldsymbol{\gamma}} \log w(X;\boldsymbol{\gamma}),$$

while a control observation contributes score $\mathbf{0}$. Hence the ordinary parametric score for the pooled data is

$$S_{\boldsymbol{\gamma}}(X, \delta) = \delta\, \dot{\ell}_{\boldsymbol{\gamma}}(X). \tag{7.5}$$

The *efficient score* $S_{\boldsymbol{\gamma}}^*$ is the residual of $S_{\boldsymbol{\gamma}}$ after projecting onto the orthogonal complement of $\mathscr{T}$ in $L_2^0(P)$. Equivalently, $S_{\boldsymbol{\gamma}}^*$ is the unique element of $L_2^0(P)$ that can be written as

$$S_{\boldsymbol{\gamma}}^*(X, \delta) = \delta\, \psi_1(X) - (1 - \delta)\, \psi_0(X) \tag{7.6}$$

for some functions ψ_1, ψ_0 satisfying $\mathbb{E}_{G_1}[\psi_1(X)] = \mathbf{0}$ and $\mathbb{E}_{G_0}[\psi_0(X)] = \mathbf{0}$, and such that $S_{\boldsymbol{\gamma}}^* \perp \mathscr{T}$.

Rather than computing the projection directly, we construct the efficient score explicitly and verify orthogonality to $\mathscr{T}$. Define the *retrospective* probability

$$\pi(x;\boldsymbol{\gamma}) = \frac{\rho\, w(x;\boldsymbol{\gamma})}{1 + \rho\, w(x;\boldsymbol{\gamma})}, \qquad \rho \equiv \frac{n_1}{n_0}. \tag{7.7}$$

Theorem 20 (Efficient score for the general DRM)**.** *The efficient score for $\boldsymbol{\gamma}$ in the semiparametric model $\mathscr{P}$ is*

$$S_{\boldsymbol{\gamma}}^*(X, \delta) = \Big\{ \delta - \pi(X;\boldsymbol{\gamma}) \Big\}\, \dot{\ell}_{\boldsymbol{\gamma}}(X) \tag{7.8}$$

Moreover, the baseline–specific efficient score function is

$$S_{\boldsymbol{\gamma}}^*(X) = \frac{1}{1 + \rho\, w(X;\boldsymbol{\gamma})}\, \dot{\ell}_{\boldsymbol{\gamma}}(X) - \mathbb{E}_{G_0}\left[\frac{w(X;\boldsymbol{\gamma})}{1 + \rho\, w(X;\boldsymbol{\gamma})}\, \dot{\ell}_{\boldsymbol{\gamma}}(X) \right], \tag{7.9}$$

which satisfies $\mathbb{E}_{G_0}[S_{\boldsymbol{\gamma}}^*(X)] = \mathbf{0}$.

Proof. We verify that (7.8) is orthogonal to every element of the nuisance tangent space $\mathscr{T}$. Let $a(X) \in \mathscr{T}$. Using the inner product induced by the stratified sampling,

$$\langle S_{\boldsymbol{\gamma}}^*, a \rangle = \frac{n_0}{n}\, \mathbb{E}_{G_0}\big[S_{\boldsymbol{\gamma}}^*(X, 0)\, a(X)\big] + \frac{n_1}{n}\, \mathbb{E}_{G_0}\big[w(X)\, S_{\boldsymbol{\gamma}}^*(X, 1)\, a(X)\big].$$

Substituting $S^*_\gamma(X,0) = -\pi(X)\dot{\ell}_\gamma(X)$ and $S^*_\gamma(X,1) = \{1-\pi(X)\}\dot{\ell}_\gamma(X)$, we obtain

$$\begin{aligned}\langle S^*_\gamma, a\rangle &= \frac{n_0}{n}\,\mathbb{E}_{G_0}\big[-\pi(X)\dot{\ell}_\gamma(X)a(X)\big] + \frac{n_1}{n}\,\mathbb{E}_{G_0}\big[w(X)\{1-\pi(X)\}\dot{\ell}_\gamma(X)a(X)\big] \\ &= \mathbb{E}_{G_0}\left[\left\{-\frac{n_0}{n}\pi(X) + \frac{n_1}{n}w(X)\big[1-\pi(X)\big]\right\}\dot{\ell}_\gamma(X)a(X)\right].\end{aligned}$$

Recall that $\rho = n_1/n_0$, so $n_0/n = (1+\rho)^{-1}$ and $n_1/n = \rho(1+\rho)^{-1}$. Moreover,

$$\pi(X) = \frac{\rho w(X)}{1+\rho w(X)}, \qquad 1-\pi(X) = \frac{1}{1+\rho w(X)}.$$

Hence

$$-\frac{n_0}{n}\pi(X) + \frac{n_1}{n}w(X)\big[1-\pi(X)\big] = -\frac{1}{1+\rho}\cdot\frac{\rho w}{1+\rho w} + \frac{\rho}{1+\rho}\cdot\frac{w}{1+\rho w} = 0.$$

Therefore $\langle S^*_\gamma, a\rangle = 0$ for every $a \in \mathscr{T}$, establishing that S^*_γ is orthogonal to the nuisance tangent space. Because S^*_γ differs from the parametric score S_γ by a function of X alone (and such functions belong to $\mathscr{T}^\perp$), it is precisely the projection of S_γ onto $\mathscr{T}^\perp$; i.e. the efficient score.

To obtain the baseline–specific form (7.9), write

$$S^*_\gamma(X,\delta) = \delta\dot{\ell}_\gamma(X) - \pi(X)\dot{\ell}_\gamma(X).$$

Taking expectation with respect to G_1 and using $dG_1 = w\,dG_0$,

$$\mathbb{E}_{G_1}\big[S^*_\gamma(X,1)\big] = \mathbb{E}_{G_0}\big[w(X)\dot{\ell}_\gamma(X)\big] - \mathbb{E}_{G_0}\big[\pi(X)\dot{\ell}_\gamma(X)\big].$$

Rearranging yields (7.9) after centering so that $\mathbb{E}_{G_0}[S^*_\gamma(X)] = \mathbf{0}$. □

Remark 21 (Role of the sampling fraction). The efficient score (7.8) depends on the design only through the ratio $\rho = n_1/n_0$. In the asymptotic regime $n_1/n \to \rho^* \in (0,1)$, the limiting efficient score replaces ρ by $\rho^*/(1-\rho^*)$. This reflects the fact that the information contributed by each sample is weighted by its respective sampling probability.

7.4. **Semiparametric information bound.** The *semiparametric information matrix* for $\boldsymbol{\gamma}$ is the variance of the efficient score under the true distribution:

(7.10)
$$\mathscr{I}^*(\boldsymbol{\gamma}) = \mathbb{E}\big[S^*_\gamma(X,\delta)S^*_\gamma(X,\delta)^\top\big] = \frac{n_0}{n}\,\mathbb{E}_{G_0}\big[S^*_\gamma(X,0)S^*_\gamma(X,0)^\top\big] + \frac{n_1}{n}\,\mathbb{E}_{G_0}\big[w(X)S^*_\gamma(X,1)S^*_\gamma(X,1)^\top\big].$$

Theorem 22 (Semiparametric information for the general DRM)**.** *Let* $\pi(x;\boldsymbol{\gamma})$ *be defined by* (7.7)*. The semiparametric information matrix is*

$$\mathscr{I}^*(\boldsymbol{\gamma}) = \mathbb{E}_{G_0}\left[\pi(X;\boldsymbol{\gamma})\big\{1-\pi(X;\boldsymbol{\gamma})\big\}\dot{\ell}_\gamma(X)\dot{\ell}_\gamma(X)^\top\right]. \tag{7.11}$$

Proof. Using the identity $\delta^2 = \delta$ and the mixture structure of the sampling design,

$$\begin{aligned}\mathscr{I}^*(\boldsymbol{\gamma}) &= \mathbb{E}\left[\big\{\delta - \pi(X)\big\}^2\dot{\ell}_\gamma(X)\dot{\ell}_\gamma(X)^\top\right] \\ &= \mathbb{E}_{G_0}\left[\mathbb{E}\big[\{\delta-\pi(X)\}^2 \mid X\big]\,\dot{\ell}_\gamma(X)\dot{\ell}_\gamma(X)^\top\right].\end{aligned}$$

Now $\mathbb{E}[\delta \mid X] = \pi(X)$ and $\mathrm{Var}(\delta \mid X) = \pi(X)\{1-\pi(X)\}$, so $\mathbb{E}[\{\delta-\pi(X)\}^2 \mid X] = \pi(X)\{1-\pi(X)\}$. This yields (7.11). □

Corollary 23 (Efficient influence function for $G_0(x)$)**.** *The efficient influence function for estimating the baseline CDF $G_0(x)$ at a fixed point x is*

$$\psi^*_{0,x}(X,\delta) = \mathbf{1}_{\{X\le x\}} - G_0(x) - H_0(x)^\top \mathscr{I}^*(\boldsymbol{\gamma})^{-1} S^*_{\boldsymbol{\gamma}}(X,\delta), \tag{7.12}$$

where

$$H_0(x) = \mathbb{E}_{G_0}\Big[\mathbf{1}_{\{X\le x\}}\pi(X;\boldsymbol{\gamma})\,\dot{\ell}_{\boldsymbol{\gamma}}(X)\Big].$$

The asymptotic variance of any regular estimator of $G_0(x)$ is bounded below by

$$\frac{G_0(x)\{1-G_0(x)\}}{\rho} - H_0(x)^\top \mathscr{I}^*(\boldsymbol{\gamma})^{-1} H_0(x),$$

where $\rho = \lim n_0/n$.

Proof. The nonparametric influence function for $G_0(x)$ is $\mathbf{1}_{\{X\le x\}} - G_0(x)$. Its projection onto the nuisance tangent space $\mathscr{T}$ is $H_0(x)^\top \mathscr{I}^*(\boldsymbol{\gamma})^{-1} S^*_{\boldsymbol{\gamma}}(X,\delta)$, because $S^*_{\boldsymbol{\gamma}}$ spans the orthogonal complement of $\mathscr{T}$. Subtracting this projection yields the efficient influence function (7.12). The variance bound follows from $\mathbb{E}[(\psi^*_{0,x})^2]$. □

7.5. **Attainment of the bound by the MELE.** We now establish that the maximum empirical likelihood estimator $\tilde{\boldsymbol{\gamma}}$ attains the semiparametric information bound $\mathscr{I}^*(\boldsymbol{\gamma}_0)^{-1}$, and therefore is asymptotically efficient.

Theorem 24 (Semiparametric efficiency of the MELE)**.** *Let $\tilde{\boldsymbol{\gamma}}$ be the MELE defined by the profile log–likelihood (2.12) with a general DRM $w(x;\boldsymbol{\gamma})$. Under Assumptions in 7.1,*

$$\sqrt{n}(\tilde{\boldsymbol{\gamma}} - \boldsymbol{\gamma}_0) \xrightarrow{d} \mathscr{N}_{p+1}\big(\mathbf{0}, \mathscr{I}^*(\boldsymbol{\gamma}_0)^{-1}\big). \tag{7.13}$$

Consequently, $\tilde{\boldsymbol{\gamma}}$ is asymptotically efficient among all regular estimators of $\boldsymbol{\gamma}_0$ in the semiparametric model $\mathscr{P}$.

Proof. From Theorem 14 (general DRM asymptotic normality) we have

$$\sqrt{n}(\tilde{\boldsymbol{\gamma}} - \boldsymbol{\gamma}_0) \xrightarrow{d} \mathscr{N}(\mathbf{0}, \mathscr{I}_{\mathrm{prof}}(\boldsymbol{\gamma}_0)^{-1}).$$

The profile information matrix $\mathscr{I}_{\mathrm{prof}}(\boldsymbol{\gamma})$ for the general DRM is derived from the Hessian of the profile log–likelihood (2.12). Direct calculation shows that its limit in probability coincides with $\mathscr{I}^*(\boldsymbol{\gamma}_0)$ given in (7.11); see **?** for the general argument. By the convolution theorem of Hájek and Le Cam, any regular estimator has asymptotic variance at least $\mathscr{I}^*(\boldsymbol{\gamma}_0)^{-1}$. Since $\tilde{\boldsymbol{\gamma}}$ achieves exactly this variance, it is semiparametrically efficient. □

Remark 25 (Efficiency gain relative to the parametric model). The semiparametric information $\mathscr{I}^*(\boldsymbol{\gamma})$ is strictly smaller (in the positive–definite sense) than the information obtained by treating G_0 as known, because the projection onto $\mathscr{T}^\perp$ removes the component of the score attributable to uncertainty in G_0. However, $\mathscr{I}^*(\boldsymbol{\gamma})$ is larger than the information from the case sample alone, because the DRM constraint pools information across both samples. The efficiency gain is quantified by the term $H_0(x)^\top \mathscr{I}^*(\boldsymbol{\gamma})^{-1} H_0(x)$ in Corollary 23.

7.6. **Special case: the exponential tilt DRM.** When $w(x;\boldsymbol{\gamma})$ takes the exponential tilt form $w(x;\boldsymbol{\gamma}) = \exp\{\alpha + \boldsymbol{\beta}^\top \mathbf{r}(x)\}$, the general results above reduce to the classical formulae. In this case $\dot{\ell}_{\boldsymbol{\gamma}}(x) = (1, \mathbf{r}(x)^\top)^\top$, and the profile log–likelihood (2.12) simplifies to

$$\ell_{\text{prof}}(\boldsymbol{\gamma}) = \sum_{j=1}^{n_1} \log w(z_j;\boldsymbol{\gamma}) - \sum_{i=1}^{n} \log\Big\{1 + \frac{n_1}{n_0} w(t_i;\boldsymbol{\gamma})\Big\} + \text{const.} \tag{7.14}$$

Differentiating yields the profile score

$$\mathbf{U}_n(\boldsymbol{\gamma}) = \sum_{i=1}^{n} \Big\{\delta_i - \pi(t_i;\boldsymbol{\gamma})\Big\} \begin{pmatrix} 1 \\ \mathbf{r}(t_i) \end{pmatrix}, \tag{7.15}$$

which coincides with the efficient score (7.8). The semiparametric information matrix becomes

$$\mathscr{I}^*(\boldsymbol{\gamma}) = \mathbb{E}_{G_0}\left[\pi(X;\boldsymbol{\gamma})\{1 - \pi(X;\boldsymbol{\gamma})\} \begin{pmatrix} 1 \\ \mathbf{r}(X) \end{pmatrix} \begin{pmatrix} 1 \\ \mathbf{r}(X) \end{pmatrix}^\top\right],$$

and the Lagrange multiplier has the closed–form $\lambda_1 = n_1 - n_0$ (Corollary 7), which yields the explicit algebraic simplification in the proof of Theorem **??**.

7.7. **Extension to the model with known constant θ.** We now return to the original formulation (7.1) with known design constant $\theta > 0$:

$$\frac{dG_1}{dG_0}(x) = \theta\, w(x;\boldsymbol{\gamma}).$$

Because θ is known, we may absorb it into a reparameterization. Let $\boldsymbol{\gamma} = (\alpha, \boldsymbol{\beta}^\top)^\top$ and define $w_\theta(x;\boldsymbol{\gamma}) = \theta\, w(x;\boldsymbol{\gamma})$. The reparameterized model is

$$\frac{dG_1}{dG_0}(x) = w_\theta(x;\boldsymbol{\gamma}),$$

which is exactly the canonical DRM studied above with the general weight function w_θ. All preceding results apply verbatim provided w is replaced by w_θ.

Corollary 26 (Efficient score with known θ)**.** *Let $\boldsymbol{\gamma} = (\alpha, \boldsymbol{\beta}^\top)^\top$ with θ known. The efficient score for $\boldsymbol{\gamma}$ is*

$$S^*_{\boldsymbol{\gamma}}(X,\delta;\theta) = \Big\{\delta - \pi_\theta(X;\boldsymbol{\gamma})\Big\}\, \dot{\ell}_{\boldsymbol{\gamma}}(X), \tag{7.16}$$

where

$$\pi_\theta(X;\boldsymbol{\gamma}) = \frac{\rho\,\theta\, w(X;\boldsymbol{\gamma})}{1 + \rho\,\theta\, w(X;\boldsymbol{\gamma})}. \tag{7.17}$$

The semiparametric information bound is

$$\mathscr{I}^*_\theta(\boldsymbol{\gamma}) = \mathbb{E}_{G_0}\Big[\pi_\theta(X;\boldsymbol{\gamma})\Big\{1 - \pi_\theta(X;\boldsymbol{\gamma})\Big\}\, \dot{\ell}_{\boldsymbol{\gamma}}(X)\dot{\ell}_{\boldsymbol{\gamma}}(X)^\top\Big]. \tag{7.18}$$

Proof. Substitute $w_\theta(x;\boldsymbol{\gamma}) = \theta w(x;\boldsymbol{\gamma})$ into Theorem 20. The sampling ratio ρ remains unchanged because it is determined by the design, while the population odds θ enters π_θ multiplicatively. The information matrix (7.18) follows from (7.11) with π replaced by π_θ. □

Remark 27 (Interpretation). The known constant θ shifts the effective parameterization of the density ratio. In case–control studies, θ is the population control–to–case ratio; when $\theta \neq 1$, the efficient score adjusts the retrospective probabilities π_θ to reflect the true population prevalence rather than the sampling prevalence. This adjustment is crucial for valid semiparametric inference under stratified sampling.

Theorem 28 (Efficiency of the profile MELE with known θ)**.** *Let $\tilde{\gamma}_\theta$ maximize the profile log–likelihood (2.12) with $w(x;\gamma)$ replaced by $\theta w(x;\gamma)$. Then*

$$\sqrt{n}(\tilde{\gamma}_\theta - \gamma_0) \xrightarrow{d} \mathcal{N}_{p+1}\big(\mathbf{0}, \mathscr{I}_\theta^*(\gamma_0)^{-1}\big),$$

and $\tilde{\gamma}_\theta$ attains the semiparametric efficiency bound for the model with known θ.

Proof. The profile log–likelihood with known θ is identical to that of the canonical DRM after the multiplicative adjustment $w \mapsto \theta w$. Since $\mathscr{I}_\theta^*(\gamma) = \mathscr{I}^*(\gamma;\theta)$, the result follows immediately from Theorem 24. □

8. Applications to Hypothesis Testing

The asymptotic results developed in Sections 6 and 7 enable several important inferential procedures.

8.1. **Goodness-of-Fit Testing.** To test the validity of a specified density ratio function $w(x;\gamma)$, we may use the Kolmogorov–Smirnov-type statistic

$$T_n = \sup_{x\in\mathbb{R}} \sqrt{n}\big|\tilde{G}_0(x) - \mathbb{G}_{n_0}(x)\big|, \tag{8.1}$$

where $\mathbb{G}_{n_0}(x) = n_0^{-1}\sum_{i=1}^{n_0} \mathbf{1}_{\{x_i \le x\}}$ is the empirical CDF of the control sample. Under the null hypothesis that the DRM holds, Theorem 16 implies

$$T_n \xrightarrow{d} \sup_{x\in\mathbb{R}}\big|\mathbb{G}_0(x) - \mathbb{G}_0^*(x)\big|, \tag{8.2}$$

where $\mathbb{G}_0^*$ is an independent copy of the limiting Gaussian process. Critical values can be obtained via the multiplier bootstrap Kosorok [2008].

8.2. **Two-Sample Testing.** For testing $H_0 : \gamma = \gamma_0$ against $H_1 : \gamma \neq \gamma_0$, the likelihood ratio statistic is

$$\Lambda_n = 2\big\{\ell_{\text{prof}}(\tilde{\gamma}) - \ell_{\text{prof}}(\gamma_0)\big\}. \tag{8.3}$$

Theorem 29 (Wilks' Theorem)**.** *Under $H_0 : \gamma = \gamma_0$ and Assumptions 7.1–7.1,*

$$\Lambda_n \xrightarrow{d} \chi_p^2. \tag{8.4}$$

Proof. This follows from standard Taylor expansion of the profile log-likelihood around $\tilde{\gamma}$ and the asymptotic normality established in Theorem 14. □

8.3. **Score and Wald Tests.** In addition to the likelihood ratio test, the score and Wald tests provide asymptotically equivalent alternatives. The score test statistic

$$R_n = \mathbf{S}_{\text{prof}}(\gamma_0)^\top \mathscr{I}_{\text{prof}}(\gamma_0)^{-1}\mathbf{S}_{\text{prof}}(\gamma_0) \tag{8.5}$$

and the Wald test statistic

$$W_n = n(\tilde{\gamma} - \gamma_0)^\top \mathscr{I}_{\text{prof}}(\tilde{\gamma})(\tilde{\gamma} - \gamma_0) \tag{8.6}$$

both converge to χ_p^2 under H_0, forming the classical likelihood-based inference triad Rao [1973].

9. Rare Event Detection and Imbalanced Learning

A pervasive problem in modern classification is *imbalanced learning*: the target event (case) occurs with vanishingly small population prevalence $\pi_0 := P(\delta = 1) \ll 1$, while the majority class (control) dominates. Standard empirical–risk minimizers trained on a random prospective sample therefore overwhelmingly predict the majority class. Common machine–learning remedies—random undersampling of controls, synthetic oversampling (SMOTE), or heuristic cost–sensitive reweighting—are either informationally wasteful (undersampling discards data) or rest on strong, untestable parametric assumptions (SMOTE assumes local uniformity of the case density between k–nearest neighbours).

The GLM–based density ratio model offers a rigorous probabilistic alternative that requires *none* of these devices. The key is that the population prevalence π_0 is typically known from administrative records, disease registries, or domain knowledge, and enters the likelihood as the known constant

$$\theta = \frac{1-\pi_0}{\pi_0} \gg 1. \tag{9.1}$$

The model (7.1) then automatically calibrates the retrospective likelihood to the prospective population, and the semiparametric efficient estimator adapts to the rarity of the event without artificial resampling.

9.1. **Prospective risk calibration from retrospective data.** In a rare–event setting the training data are almost always collected retrospectively: the investigator fixes the number of cases n_1 and controls n_0 independently, yielding a design ratio $\rho = n_1/n_0$ that is $O(1)$ (e.g. $1:1$ or $1:5$) even though $\pi_0 \approx 0$. The density of (X, δ) under this stratified design is

$$f(x,\delta) = \left(\frac{n_0}{n}\right)^{1-\delta}\left(\frac{n_1}{n}\right)^{\delta}\big[\theta\, w(x;\boldsymbol{\gamma})\big]^{\delta} g_0(x), \qquad n = n_0 + n_1.$$

Because θ is known, we reparametrize $w_\theta(x;\boldsymbol{\gamma}) = \theta\, w(x;\boldsymbol{\gamma})$ as in Section 7. The prospective (population) risk conditional on covariates is then identified by Bayes' theorem:

$$\begin{aligned} P(\delta = 1 \mid X = x) &= \frac{\pi_0\, g_1(x)}{\pi_0\, g_1(x) + (1-\pi_0)\, g_0(x)} \\ &= \frac{\pi_0\, \theta\, w(x;\boldsymbol{\gamma})}{\pi_0\, \theta\, w(x;\boldsymbol{\gamma}) + (1-\pi_0)}. \end{aligned}$$

Since $\pi_0\theta = 1 - \pi_0$ by (9.1), this collapses to the canonical form

$$\pi(x;\boldsymbol{\gamma}) := P(\delta = 1 \mid X = x) = \frac{w(x;\boldsymbol{\gamma})}{1 + w(x;\boldsymbol{\gamma})}. \tag{9.2}$$

Thus the population risk depends on $\boldsymbol{\gamma}$ *only* through the density ratio $w(x;\boldsymbol{\gamma})$; the extreme rarity π_0 has been absorbed into the intercept by the normalization constraint $\int w_\theta\, dG_0 = 1$. The plug–in estimator

$$\hat{\pi}(x) = \frac{w(x;\tilde{\boldsymbol{\gamma}}_\theta)}{1 + w(x;\tilde{\boldsymbol{\gamma}}_\theta)} \tag{9.3}$$

is uniformly consistent and asymptotically linear, with an influence function derived from the efficient score.

Theorem 30 (Efficient prospective risk estimation)**.** *Let $\pi_0 \in (0,1)$ be known with $\theta = (1-\pi_0)/\pi_0$, and let $\tilde{\boldsymbol{\gamma}}_\theta$ be the profile MELE of Section 7. Under Assumptions 7.1–7.1,*

$$\sup_{x\in\mathscr{D}}\big|\hat{\pi}(x) - \pi(x)\big| \xrightarrow{p} 0. \tag{9.4}$$

Moreover, for any fixed x,

$$\sqrt{n}\big\{\hat{\pi}(x) - \pi(x)\big\} = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_\pi(X_i,\delta_i) + o_p(1), \tag{9.5}$$

with influence function

$$\psi_\pi(X,\delta) = \frac{\dot{\ell}_{\boldsymbol{\gamma}}(x)^\top}{\{1+w(x;\boldsymbol{\gamma}_0)\}^2}\,\mathscr{I}_\theta^*(\boldsymbol{\gamma}_0)^{-1}S_{\boldsymbol{\gamma}}^*(X,\delta;\theta), \tag{9.6}$$

where $S_{\boldsymbol{\gamma}}^*(X,\delta;\theta)$ is the efficient score (7.16). Consequently $\sqrt{n}\{\hat{\pi}(x) - \pi(x)\} \xrightarrow{d} \mathscr{N}(0,\sigma_\pi^2(x))$ with $\sigma_\pi^2(x) = \mathbb{E}[\psi_\pi^2]$, and this variance attains the semiparametric efficiency bound for estimating the prospective risk $\pi(x)$.

Proof. By Theorem 28, $\tilde{\boldsymbol{\gamma}}_\theta$ is asymptotically linear with influence function $\mathscr{I}_\theta^*(\boldsymbol{\gamma}_0)^{-1}S_{\boldsymbol{\gamma}}^*(X,\delta;\theta)$. The map $\boldsymbol{\gamma} \mapsto w(x;\boldsymbol{\gamma})/\{1+w(x;\boldsymbol{\gamma})\}$ is smooth with gradient $\dot{\ell}_{\boldsymbol{\gamma}}(x)/\{1+w(x;\boldsymbol{\gamma})\}^2$; the delta method yields (9.6). Uniform consistency follows from Assumption 7.1 and the Donsker property (Assumption 7.1), which guarantees stochastic equicontinuity of the empirical process indexed by $\boldsymbol{\gamma} \in \Theta$. □

9.2. **Why resampling is unnecessary and inefficient.**

The likelihood self–corrects for imbalance. The efficient score for the rare–event DRM is

$$S_{\boldsymbol{\gamma}}^*(X,\delta;\theta) = \big\{\delta - \pi_\theta(X;\boldsymbol{\gamma})\big\}\,\dot{\ell}_{\boldsymbol{\gamma}}(X), \qquad \pi_\theta(X;\boldsymbol{\gamma}) = \frac{\rho\,\theta\,w(X;\boldsymbol{\gamma})}{1+\rho\,\theta\,w(X;\boldsymbol{\gamma})}. \tag{9.7}$$

Because the training sample is enriched with cases, the design ratio $\rho = n_1/n_0$ is $O(1)$; because the event is rare in the population, $\theta \gg 1$. The product $\rho\theta\,w(X;\boldsymbol{\gamma}) = \rho\,w_\theta(X;\boldsymbol{\gamma})$ depends on the *population* density ratio $w_\theta = \theta w$, which satisfies the normalization $\int w_\theta\,dG_0 = 1$ and is therefore $O(1)$. Consequently $\pi_\theta(X;\boldsymbol{\gamma})$ remains bounded away from 0 and 1 (Assumption 7.1 applied to w_θ), and the information matrix

$$\mathscr{I}_\theta^*(\boldsymbol{\gamma}) = \mathbb{E}_{G_0}\Big[\pi_\theta(X;\boldsymbol{\gamma})\{1-\pi_\theta(X;\boldsymbol{\gamma})\}\,\dot{\ell}_{\boldsymbol{\gamma}}(X)\dot{\ell}_{\boldsymbol{\gamma}}(X)^\top\Big] \tag{9.8}$$

is well–conditioned. The total information grows linearly with the total sample size $n = n_0 + n_1$; every control observation contributes to estimating G_0, and every case observation contributes to estimating the density ratio. No data are discarded and no synthetic observations are manufactured. Undersampling inflates variance. Random undersampling reduces the effective sample size to $2n_1 \ll n$ (or some other fixed ratio), discarding the vast majority of control observations. The asymptotic variance of any estimator based on the undersampled data is inflated by a factor of $n/(2n_1)$ relative to the full–sample MELE. Since the MELE already achieves the semiparametric efficiency bound, undersampling is strictly suboptimal.

SMOTE imposes an untestable density assumption. SMOTE generates synthetic case observations by convex interpolation between a case and its k–nearest case neighbours. Implicitly, this assumes that the case density g_1 is locally uniform on the line segments connecting neighbours, i.e. that g_1 is linear (or at least Lipschitz with a very small constant) in regions where the data are sparse. This is a strong parametric assumption that is rarely justified and that cannot be tested from the data at hand. When the assumption fails—as it typically does in high dimensions or near decision boundaries—SMOTE biases the score function and corrupts the information matrix, leading to inconsistent risk estimates.

By contrast, the DRM likelihood uses only the real observations (X_i, δ_i) and corrects for the retrospective design through the exact multiplicative factor θ, which is known from population–level information.

9.3. **Semiparametric efficiency under extreme imbalance.** A remarkable feature of the rare–event DRM is that the semiparametric efficiency bound does *not* degenerate as the population prevalence $\pi_0 \to 0$.

Theorem 31 (Efficiency under extreme imbalance)**.** *Let* $\{\pi_0^{(m)}\}_{m\geq 1}$ *be a sequence of population prevalences with* $\pi_0^{(m)} \to 0$*, and let* $\theta^{(m)} = (1-\pi_0^{(m)})/\pi_0^{(m)} \to \infty$*. Suppose the design ratio* $\rho = n_1/n_0$ *is held fixed and that Assumption 7.1 holds uniformly for the sequence* $w_{\theta^{(m)}} = \theta^{(m)} w$*. Then the semiparametric information matrix* $\mathscr{I}^*_{\theta^{(m)}}(\boldsymbol{\gamma}_0)$ *satisfies*

$$0 < \liminf_{m\to\infty} \lambda_{\min}\big(\mathscr{I}^*_{\theta^{(m)}}(\boldsymbol{\gamma}_0)\big) \leq \limsup_{m\to\infty} \lambda_{\max}\big(\mathscr{I}^*_{\theta^{(m)}}(\boldsymbol{\gamma}_0)\big) < \infty. \tag{9.9}$$

Moreover, the asymptotic variance of the MELE $\tilde{\boldsymbol{\gamma}}_{\theta^{(m)}}$ *is* $O(1/n)$ *uniformly in* m*:*

$$\sup_m \left\|\mathrm{Var}\big(\sqrt{n}\,(\tilde{\boldsymbol{\gamma}}_{\theta^{(m)}} - \boldsymbol{\gamma}_0)\big)\right\| < \infty. \tag{9.10}$$

Proof. By Assumption 7.1 applied to $w_{\theta^{(m)}}$, there exist constants $0 < c_1 \leq c_2 < \infty$ such that $c_1 \leq w_{\theta^{(m)}}(x) \leq c_2$ for all x and all m. Hence

$$\frac{\rho c_1}{1+\rho c_1} \leq \pi_{\theta^{(m)}}(x) \leq \frac{\rho c_2}{1+\rho c_2},$$

so that $\pi_{\theta^{(m)}}(x)\{1-\pi_{\theta^{(m)}}(x)\}$ is bounded away from 0 uniformly in m. The information matrix (9.8) is therefore a uniform expectation of a positive–definite kernel weighted by a uniformly bounded, uniformly positive function. By Assumption 7.1, the smallest eigenvalue is bounded away from 0, yielding (9.9). The variance bound (9.10) follows from Theorem 28 and the uniform boundedness of $\mathscr{I}^*_{\theta^{(m)}}(\boldsymbol{\gamma}_0)^{-1}$. □

Remark 32 (Implications for practice). Theorem 31 shows that the DRM estimator remains efficient even when the event is extremely rare (e.g. $\pi_0 = 10^{-4}$ or 10^{-6}), provided the total sample size n is large enough for the asymptotic approximations to hold. The population rarity π_0 does *not* appear in the limiting variance; only the design ratio ρ and the population density ratio w_θ matter. This is in sharp contrast to prospective estimators that rely on observing the rare event in a random sample, whose variance scales as $1/(n\pi_0)$ and therefore explodes as $\pi_0 \to 0$.

9.4. **Decision–theoretic implications.** In imbalanced classification the loss function is often asymmetric: a false negative (missing a case) is far more costly than a false positive. Let $C_{01} > 0$ be the cost of misclassifying a control as a case and $C_{10} > 0$ the cost of misclassifying a case as a control. The Bayes–optimal decision rule is

$$\hat{\delta}(x) = \mathbf{1}_{\{\hat{\pi}(x)>\tau\}}, \qquad \tau = \frac{C_{01}}{C_{01}+C_{10}}.$$

Because $\hat{\pi}(x)$ is efficiently estimated, the plug–in rule attains the minimum expected cost among all regular classifiers based on the same covariates. Moreover, because $\hat{\pi}(x)$ is calibrated to the true population prevalence via (9.2), the threshold τ does not require ad–hoc adjustment (unlike the heuristic "balance the training set" approach, which implicitly assumes $\tau = 1/2$).

Corollary 33 (Efficiency of the plug–in Bayes classifier)**.** *Let $\hat{\delta}_n(x) = \mathbf{1}_{\{\hat{\pi}(x)>\tau\}}$ be the plug–in classifier with $\hat{\pi}$ from (9.3). Under the assumptions of Theorem 30, the expected cost $\mathscr{R}(\hat{\delta}_n) = \mathbb{E}[C_{01}(1-\delta)\hat{\delta}_n(X) + C_{10}\delta(1-\hat{\delta}_n(X))]$ satisfies*

$$\mathscr{R}(\hat{\delta}_n) - \mathscr{R}^* = O\big(n^{-1/2}\big),$$

where $\mathscr{R}^*$ is the Bayes risk.

Proof. The difference in risks is controlled by the L_1–error $\mathbb{E}\,|\hat{\pi}(X) - \pi(X)|$, which is $O(n^{-1/2})$ by Theorem 30 and the Cauchy–Schwarz inequality. □

9.5. **Bayesian connections and the semiparametric Bernstein–von Mises theorem.** Although the DRM is developed within a frequentist semiparametric framework, its profile empirical likelihood admits a natural Bayesian interpretation. The empirical likelihood (2.12) is a nonparametric likelihood in the sense of Owen [1988, 2001]: it assigns multinomial weights p_i to the observed atoms and profiles out the infinite–dimensional nuisance G_0 by constrained maximisation. This profiling operation is the frequentist analogue of marginalizing over a nonparametric prior on G_0.

Specifically, suppose one places a prior $\Pi(\boldsymbol{\gamma})$ on the finite–dimensional parameter and a nonparametric prior $\Pi(G_0)$ (e.g. a Dirichlet process with concentration parameter $\alpha \to 0$) on the baseline distribution (See Ferguson [1973], Antoniak [1974]). The marginal posterior for $\boldsymbol{\gamma}$ is then proportional to

$$\Pi(\boldsymbol{\gamma}) \times \int \prod_{i=1}^{n} p_i(\boldsymbol{\gamma}, G_0)\, d\Pi(G_0),$$

where $p_i(\boldsymbol{\gamma}, G_0)$ are the observation probabilities under the DRM constraint. Schennach [2005] showed that, under mild conditions, the exponentially tilted empirical likelihood is exactly the marginal likelihood obtained after integrating out a nonparametric latent distribution with a suitable prior; see also Lazar [2003] and Newton and Raftery [1994] for earlier connections between empirical likelihood and Bayesian inference via the weighted likelihood bootstrap.

In the rare–event setting, the known population prevalence π_0 (or equivalently θ) plays the role of an informative hyperparameter. From a Bayesian perspective, θ supplies prior information about the relative frequency of the two strata that is independent of the covariates, thereby strongly regularizing the posterior for the prospective risk $\pi(x)$ even when the case sample n_1 is small. This explains why the DRM approach avoids the degeneracy that plagues likelihood–based classifiers trained on imbalanced data: the prior information encoded in θ prevents the posterior from collapsing to the trivial majority–class predictor.

The semiparametric information bound $\mathscr{I}^*_\theta(\boldsymbol{\gamma}_0)^{-1}$ derived in Theorem 22 has a direct Bayesian counterpart. Under a nonparametric prior on G_0 that satisfies the conditions of the semiparametric Bernstein–von Mises theorem (Bickel and Kleijn [2012]), the posterior for $\boldsymbol{\gamma}$ concentrates at rate $n^{-1/2}$ and is asymptotically normal with covariance matrix $\mathscr{I}^*_\theta(\boldsymbol{\gamma}_0)^{-1}$. Consequently, credible sets constructed from the posterior coincide asymptotically with the efficient confidence intervals based on the MELE, and the posterior mean (or mode) is asymptotically equivalent to $\tilde{\boldsymbol{\gamma}}_\theta$. In this sense the MELE can be viewed as the limit of a Bayesian posterior summary under a diffuse nonparametric prior, and the efficient score (7.16) is the gradient of the log–posterior density in the limit experiment.

10. Application Domains and Open Directions

The density ratio model (7.1) is fundamentally a semiparametric framework for comparing two populations. Consequently, every scientific or engineering problem that can be cast as a contrast between a reference distribution G_0 and a perturbed distribution G_1 is, in principle, amenable to DRM analysis. To date, the methodology has been developed most intensively in two broad areas: biostatistics and medical diagnostics, and econometrics.

10.1. **Clinical studies and medical diagnostics.** In medical diagnostics, the DRM provides a natural model for evaluating the accuracy of a continuous biomarker or diagnostic test. Let G_0 be the distribution of the test result in healthy controls and G_1 its distribution in diseased cases. The ROC curve, which plots the true–positive rate $\bar{G}_1(c)$ against the false–positive rate $\bar{G}_0(c)$ as the threshold c varies, is entirely determined by the density ratio $w(x; \boldsymbol{\gamma}) = dG_1/dG_0$. Under an exponential tilt, the ROC curve has a closed–form logistic expression, and the AUC (equivalently the Gini coefficient) can be estimated semiparametrically without fully specifying either G_0 or G_1 as in Qin and Zhang [2003]. The Youden index, which maximizes the sum of sensitivity and specificity minus one, and the optimal threshold derived therefrom, are likewise efficiently estimated under the DRM in Yuan et al. [2021]. More generally, case–control studies with known sampling fractions use the DRM to pool information across strata, yielding efficient estimators of odds ratios and attributable risk while treating the baseline disease distribution nonparametrically (see Qin [1998]).

10.2. **Economics and econometrics.** In program evaluation and causal inference, the DRM arises naturally when the distribution of covariates differs between a treated group (G_1) and a control group (G_0). The density ratio $w(x; \boldsymbol{\gamma}) = dG_1/dG_0$ is the propensity score reweighting function needed to correct for selection bias or covariate shift. Details is refered to Graham et al. [2012] and Imai and van Dyk [2004]. Exponential tilts are particularly convenient because they preserve the support of the covariate distribution while shifting its moments. In choice modeling and revealed–preference analysis, the DRM links the distribution of observed attributes among choosers to that among non–choosers, with the tilt parameters interpreted as marginal utilities as discussed in Cosslett [1981]. More recently, Barrett and Donald [2003] DRMs used to construct efficient estimators of treatment effect distributions and to test for stochastic dominance between income or welfare distributions without parametric assumptions on the baseline.

Despite this rich methodological development in biomedicine and econometrics, the same structural assumptions unlock powerful inference in several other domains where two–sample comparisons are ubiquitous but DRMs remain largely unexplored. Below we survey these underdeveloped areas and, in each case, provide a concrete use case that illustrates how the DRM could be deployed.

10.3. **Quality and process control.** In statistical process control (SPC), a manufacturing process is monitored by comparing a baseline "in–control" distribution G_0 (historical Phase I data) against a current "out–of–control" distribution G_1 (Phase II observations). The DRM $g_1(x) = w(x; \boldsymbol{\gamma})g_0(x)$ provides a parsimonious, interpretable model for the perturbation: the parameter $\boldsymbol{\gamma}$ describes not merely whether a shift has occurred, but *how* the shape of the distribution has changed (e.g., a tilt in the mean, variance, or tail behaviour). This is considerably richer than the traditional location–scale or moment–based control chart.

Use case: semiconductor wafer thickness monitoring. A fabrication plant maintains a baseline sample of $n_0 = 500$ wafer thickness measurements from a process known to be in control. During production, $n_1 = 50$ wafers are sampled hourly. Rather than monitoring only the sample mean (which misses variance or skewness shifts), the plant fits a DRM with $w(x; \boldsymbol{\gamma}) = \exp\{\alpha + \beta_1 x + \beta_2 x^2\}$. The estimated $\hat{\beta}_2$ detects a quadratic tilt indicating that the process variability has increased even though the mean has not shifted. Because G_0 is estimated nonparametrically from the large Phase I sample, no parametric assumption (e.g. normality) is imposed on the thickness distribution, and the semiparametric efficiency bound of Theorem 22 yields the most powerful test for the variance shift among all regular procedures.

Moreover, process capability indices such as C_{pk} and C_{pm} require estimation of tail probabilities relative to specification limits. When two production lines or supplier lots are compared, the relative capability index can be framed as a contrast between G_0 and G_1. The semiparametric efficiency theory of Section 7 yields optimal estimators of these indices without fully parametric assumptions on the process distribution, and the rare–event calibration of Section 9 is directly relevant when defect rates are measured in parts per million.

10.4. **Reliability engineering and survival analysis.** Reliability analysis is replete with two–population comparisons: accelerated life testing (ALT) compares failure–time distributions under nominal stress (G_0) and elevated stress (G_1); stress–strength models compare the distribution of strength $X \sim G_0$ to the distribution of stress $Y \sim G_1$ via the reliability $R = P(X > Y)$. In both settings, the DRM offers a flexible alternative to fully parametric (Weibull, log–normal) or proportional–hazards assumptions.

Use case: turbine blade stress–strength inference. In aerospace reliability, the tensile strength X of a turbine blade follows a baseline distribution G_0 estimated from destructive tests on $n_0 = 200$ nominally manufactured blades. The stress Y imposed during emergency overspeed conditions is modelled as G_1 with a density ratio tilt $w(y; \boldsymbol{\gamma}) = \exp\{\alpha + \beta y\}$. The system reliability $R = P(X > Y)$ is estimated under the DRM by

$$\hat{R} = \int_{-\infty}^{\infty} \hat{G}_0(y)\, d\hat{G}_1(y) = \sum_{i=1}^{n} \widetilde{G}_0(y_{(i)}) \left[\widetilde{G}_1(y_{(i)}) - \widetilde{G}_1(y_{(i-1)})\right],$$

where $\hat{G}_0$ is the empirical likelihood estimator of the baseline CDF. Because the DRM pools the strength and stress data through the shared G_0, the resulting confidence interval for R is narrower than the standard nonparametric Mann–Whitney interval, which does not exploit the tilt structure. Censored observations—common when tests are terminated at a fixed censoring time—can be incorporated by replacing the empirical likelihood with its Kaplan–Meier–weighted analogue, an extension that remains largely undeveloped in the DRM literature.

10.5. **Environmental and atmospheric science.** Environmental monitoring naturally involves before–after or control–impact comparisons: pollutant concentrations before and after a regulatory intervention, climate model output versus observed measurements, or background versus contaminated site measurements. The DRM can serve as a semiparametric bias–correction tool: $w(x; \boldsymbol{\gamma})$ tilts a climate model's predictive distribution G_0 to match the observed distribution G_1, while the baseline G_0 is estimated nonparametrically from the large model ensemble.

Use case: particulate matter regulatory assessment. A regional environmental agency compares daily $PM_{2.5}$ concentrations from a chemistry–transport model ($n_0 = 10\,000$ simulated days) against monitored observations ($n_1 = 365$ days). The model systematically underpredicts high–pollution episodes. Fitting a DRM with $w(x; \boldsymbol{\gamma}) = \exp\{\alpha + \beta \log x\}$ reveals a positive $\hat{\beta}$, indicating that the model's right tail is too light relative to reality. The tilted baseline $d\hat{G}_1 = \hat{w}\, d\hat{G}_0$ provides a calibrated forecast distribution for health–impact calculations, and the profile empirical likelihood ratio yields a confidence interval for the exceedance probability $P(X > 35\ \mu\text{g/m}^3)$ that is valid without assuming log–normality of $PM_{2.5}$ concentrations.

10.6. **Forensic science and chemometrics.** Forensic comparisons often reduce to assessing whether two samples share a common source: the elemental composition of glass fragments, the chemical profile of a controlled substance, or the spectral signature of a material. The null hypothesis of a common source is $w(x; \boldsymbol{\gamma}) \equiv 1$ (i.e. $\boldsymbol{\gamma} = \mathbf{0}$), while the alternative posits a non–trivial tilt between the reference and questioned distributions.
Use case: controlled substance profiling. Forensic chemists measure the chromatographic peak heights of $X = (X_1, \ldots, X_p)$ from a known batch of seized heroin ($n_0 = 150$ samples) and a questioned street sample ($n_1 = 10$ samples). The DRM with multivariate exponential tilt $w(x; \boldsymbol{\gamma}) = \exp\{\alpha + \boldsymbol{\beta}^\top x\}$ tests $H_0 \colon \boldsymbol{\beta} = \mathbf{0}$ against $H_1 \colon \boldsymbol{\beta} \neq \mathbf{0}$. The profile empirical likelihood ratio statistic provides a likelihood–ratio measure of evidential weight (the "weight of evidence") that avoids the strong multivariate normality assumptions underlying Hotelling's T^2. Because the baseline G_0 is estimated from the large reference library, the procedure is semiparametric efficient and robust to the heavy tails and skewness typical of chromatographic data.

Similarly, in process analytical technology (PAT), spectroscopic distributions from a reference batch (G_0) are compared against production batches (G_1) to detect deviations in chemical composition. The DRM tilt parameters can be monitored as multivariate quality indicators in real time.

10.7. **Psychometrics and educational testing.** Differential item functioning (DIF) examines whether a test item behaves differently across two groups (e.g., gender or ethnicity) after conditioning on ability. Let G_0 and G_1 denote the item–response distributions of the reference and focal groups at a given ability level. The DRM provides a semiparametric model for DIF that does not require a full item–response theory (IRT) parametrisation of the ability distribution; the tilt parameter $\boldsymbol{\gamma}$ captures uniform or non–uniform DIF through the moments of the conditional response distribution.
Use case: large–scale educational assessment. A standardised mathematics test is administered to $n_0 = 5\,000$ male students and $n_1 = 5\,000$ female students matched on total score. For a particular item, the response time X is modelled with DRM $w(x; \boldsymbol{\gamma}) = \exp\{\alpha + \beta_1 x + \beta_2 x^2\}$. A significant $\hat{\beta}_2 \neq 0$ indicates non–uniform DIF: female students with the same total score as males have a different *variability* in response time, suggesting that the item engages different cognitive processes across groups. Because G_0 is left nonparametric, the test is robust to the typically skewed, multi–modal distribution of response times. Extending DRM methodology to hierarchical item–response data is a promising direction.

10.8. **Genomics and high–throughput biology.** In differential expression analysis, the distributions of gene–level expression (or protein abundance) under condition A (G_0) and condition B (G_1) are compared for thousands of features simultaneously. While current practice relies heavily on parametric negative–binomial or log–normal models, the DRM offers a semiparametric alternative in which the baseline distribution is estimated from the data and the tilt captures condition–specific changes.
Use case: single–cell RNA–seq differential expression. A single–cell study measures the expression of a surface marker in $n_0 = 3\,000$ T–cells from healthy donors and $n_1 = 2\,000$ T–cells from autoimmune patients. The marker distribution is heavily zero–inflated and right–skewed, violating parametric assumptions. The DRM with $w(x; \boldsymbol{\gamma}) = \exp\{\alpha + \beta \log(1 + x)\}$ models a multiplicative

shift in the log–transformed abundance. The profile MELE $\hat{\beta}$ tests for differential expression while the nonparametric baseline $\hat{G}_0$ adapts automatically to the excess zeros and heavy tail. Because the baseline is shared across features (or estimated feature–wise), empirical–Bayes or hierarchical extensions of the DRM could improve power while preserving robustness to distributional skewness and heavy tails common in sequencing data.

10.9. **Covariate shift and transfer learning.** In machine learning, *covariate shift* occurs when the training distribution G_0 and the test (target) distribution G_1 differ, but the conditional outcome model remains invariant. The density ratio $w(x;\boldsymbol{\gamma}) = dG_1/dG_0$ is precisely the reweighting function needed to correct the training loss. DRMs provide a structured, semiparametric estimator of this ratio that avoids the instability of fully nonparametric kernel–based importance–weighting methods, particularly in moderate dimensions.
Use case: predictive maintenance across factories. A company trains a failure–prediction model on $n_0 = 50\,000$ sensor records from Factory A (reference distribution G_0) and wishes to deploy it in Factory B, where only $n_1 = 2\,000$ records are available (target distribution G_1). The factories differ in ambient temperature and machine age, causing covariate shift. Rather than discarding the Factory A data (undersampling) or synthesising Factory B records (SMOTE), the company fits a DRM with $w(x;\boldsymbol{\gamma}) = \exp\{\alpha + \boldsymbol{\beta}^\top x\}$ to the pooled feature vectors. The estimated weights $\hat{w}(x_i;\tilde{\boldsymbol{\gamma}})$ are used to reweight the Factory A training loss, yielding a classifier that is calibrated to Factory B's operating conditions without parametric assumptions on the high–dimensional sensor distribution. The semiparametric efficiency bound ensures that the reweighting estimator is optimal among all regular procedures that use the same features.

10.10. **Concluding remark.** The common thread across all these domains is the need to compare two populations without committing to a fully parametric model for either. The DRM achieves this by extracting a low–dimensional parametric structure from the density ratio while leaving the baseline distribution nonparametric. We anticipate that the semiparametric efficiency theory developed in this paper—particularly the calibration for rare events in Section 9—will prove equally valuable in engineering and physical sciences, where baseline distributions are often irregular and the events of interest are, by design, rare.

11. Conclusion and Future Directions

We have presented a comprehensive theoretical and methodological framework for generalized semiparametric density ratio models, unifying model formulation, computation, asymptotic theory, and practical inference. Our principal contributions are as follows.

First, we formulated a general DRM that accommodates arbitrary positive weight functions subject only to normalization constraints, substantially extending the classical exponential tilt specification. We derived the maximum empirical likelihood estimators for both the finite-dimensional parameter $\boldsymbol{\gamma}$ and the infinite-dimensional baseline distribution G_0, establishing well-posedness and consistency under standard regularity conditions.

Second, we developed three complementary computational algorithms—nested optimization, joint Newton–Raphson, and an EM-type fixed-point iteration—each equipped with rigorous convergence guarantees. The nested approach offers robustness for general weight functions; the joint Newton–Raphson achieves quadratic convergence when Hessian information is available; and the EM-type iteration provides stability for poor initial values or irregular likelihood surfaces. For the exponential tilt model, we derived a closed-form expression for the Lagrange multiplier that eliminates inner root-finding and reduces computational complexity by an order of magnitude.

Third, we established the precise equivalence between the exponential tilt DRM and multinomial logistic regression within the GLM framework. This result is of considerable practical significance, as it enables practitioners to fit semiparametric DRMs using standard statistical software while retaining the asymptotic efficiency gains of the empirical likelihood approach.

Fourth, we extended the framework to $K \geq 2$ populations, proving existence and uniqueness of the MELEs, deriving the profile score and information matrix, and establishing asymptotic normality under multi-sample sampling designs. The multi-population theory encompasses multinomial logistic regression as a special case and provides a foundation for multi-class classification and multi-arm clinical trial applications.

Fifth, we developed a complete asymptotic theory for the two-sample case, including: (i) asymptotic normality of the parameter estimator with a consistently estimable variance; (ii) weak convergence of the estimated cumulative distribution functions to tight Gaussian processes with explicit covariance structure; (iii) characterization of the semiparametric efficiency bound; and (iv) Wilks-type theorems for likelihood ratio, score, and Wald tests.

Sixth, we demonstrated how the DRM framework extends naturally to contemporary machine learning challenges, particularly rare-event detection and imbalanced learning. By incorporating known population prevalence through the design constant θ, the model self-corrects for retrospective sampling without artificial resampling, and remains semiparametrically efficient even under extreme imbalance.

Several promising directions merit further investigation. The extension to high-dimensional settings, where the dimension p of the tilt parameter grows with sample size, would broaden the applicability of DRMs to modern genomic and imaging data. Developing adaptive basis function selection procedures—potentially through penalized empirical likelihood or Bayesian nonparametric approaches—could enhance model flexibility while maintaining interpretability. Incorporating DRMs into causal inference frameworks, particularly for observational studies with complex sampling designs, represents another important avenue. Finally, deriving finite-sample refinements, such as higher-order asymptotic expansions or bootstrap calibration methods, could improve the accuracy of inference in small-sample regimes.

The density ratio model strikes a compelling balance between model flexibility and inferential efficiency. As the demand for robust, data-efficient statistical methods continues to grow across the sciences, we anticipate that the theoretical foundations and computational tools developed herein will facilitate broader adoption of DRM-based methodologies in both classical statistical applications and contemporary machine learning contexts.

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